

MA20225 — Probability 2B — Lecture Notes
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Contents

1	Discrete time Markov chains	5
1.1	What is a Markov chain?	5
1.1.1	Examples	6
1.1.2	Specification of a discrete-time Markov chain	7
1.1.3	Examples	8
1.2	m -step transition probabilities	10
1.2.1	Calculation of matrix powers by diagonalization	11
1.2.2	Questions about the long term behaviour	13
1.2.3	Tricks for Diagonalization	14
1.2.4	The method of generating functions	17
1.3	Hitting probabilities and expected waiting times	19
1.3.1	Hitting probabilities	20
1.3.2	Expected hitting times	25
1.4	Classification of states and convergence to equilibrium	29
1.4.1	Communicating classes	30
1.4.2	Period	32
1.5	Invariant distributions and convergence to equilibrium	33
1.5.1	Positive Recurrence	33
1.5.2	Convergence to equilibrium	35
1.6	Time-reversible Markov chains	41

Chapter 1

Discrete time Markov chains

In this course we consider a class of stochastic processes called Markov chains. The course is roughly equally divided between discrete-time and continuous-time Markov chains. We shall study various methods to help understand the behaviour of Markov chains, in particular over the long term.

1.1 What is a Markov chain?

A stochastic process is a mathematical model for a random evolution in time. Formally, a *discrete-time stochastic process* is a sequence $\{X_n : n = 0, 1, 2, \dots\}$ of random variables. The value of the random variable X_n is interpreted as the state of the random development after n time steps. In this unit $\{X_n\}_{n \geq 0}$ will be discrete random variables, taking values in a fixed countable set I , called the **state space**. That is, I is either finite or countably infinite, also referred to as a discrete set.

Definition 1.1. A sequence of random variables $X_0, X_1, X_2, \dots$ taking values on the discrete set I is called a **Markov chain** if for all $n \geq 0$, and all $i_0, \dots, i_{n+1} \in I$ we have

$$\mathbb{P}(X_{n+1} = i_{n+1} \mid X_n = i_n, \dots, X_1 = i_1, X_0 = i_0) = \mathbb{P}(X_{n+1} = i_{n+1} \mid X_n = i_n), \quad (**)$$

whenever the event in the conditioning has positive probability. The Markov chain is called **time-homogeneous**, if the right hand side only depends on the states i_n, i_{n+1} and not on n . In other words, if there exists a matrix P indexed by I , that does not depend on n , such that

$$\mathbb{P}(X_{n+1} = i_{n+1} \mid X_n = i_n) = P_{i_n i_{n+1}}, \quad n \geq 0.$$

In this unit, all Markov chains will be time-homogeneous.

A sequence satisfying equation (**), is said to have the *Markov property*. The implication of the Markov property is that the most recently observed value of the chain in itself is enough to describe the future values of the chain. This can be interpreted as *given the present, the future is independent of the past*.

We state below an alternative definition that formally requires more, but can be shown to be equivalent to (**).

Definition 1.2. A sequence of random variables $X_0, X_1, X_2, \dots$ taking values on the discrete set I is called a **Markov chain** if for all $n \geq 1$, indices $k_1 < k_2 < \dots < k_n < k_{n+1}$ and all $x_1, x_2, \dots, x_n, x_{n+1} \in I$,

$$\mathbb{P}(X_{k_{n+1}} = x_{n+1} | X_{k_n} = x_n, \dots, X_{k_2} = x_2, X_{k_1} = x_1) = \mathbb{P}(X_{k_{n+1}} = x_{n+1} | X_{k_n} = x_n), \quad (*)$$

whenever the conditioning has positive probability. The Markov chain is called time-homogeneous, if the right hand side only depends on x_{n+1}, x_n , and $k_{n+1} - k_n$.

A crucial first point to understand about Markov chains is that typically X_n and X_{n+1} are *not independent* (see Examples below). Therefore, you cannot directly apply results you learned about independent sequences, and a new theory is needed. The importance of Markov chains lies in the fact that in a way they represent the *simplest* kind of non-independence one can have. Their role in stochastic modelling is somewhat parallel to the role of differential equations in applied mathematics, making them a powerful tool.

1.1.1 Examples

Example 1.3. Suppose a virus can exist in two different strains α, β and in each generation either stays the same, or with probability $p \ll 1/2$ mutates to the other strain. Suppose the virus is in strain α initially, what is the probability that it is in the same strain after n generations?

We let X_n be the strain of the virus in the n th generation, which is a random variable with values in $I = \{\alpha, \beta\}$. We show that X_n and X_{n+1} are not independent. First, we compute

$$\mathbb{P}\{X_n = \alpha, X_{n+1} = \alpha\} = \mathbb{P}\{X_n = \alpha \text{ and no mutation occurs in step } n+1\} = \mathbb{P}\{X_n = \alpha\} (1 - p),$$

$$\mathbb{P}\{X_n = \beta, X_{n+1} = \alpha\} = \mathbb{P}\{X_n = \beta \text{ and mutation occurs in step } n+1\} = \mathbb{P}\{X_n = \beta\} p,$$

and hence

$$\begin{aligned} \mathbb{P}\{X_{n+1} = \alpha\} &= \mathbb{P}\{X_n = \alpha, X_{n+1} = \alpha\} + \mathbb{P}\{X_n = \beta, X_{n+1} = \alpha\} \\ &= (1 - p)\mathbb{P}\{X_n = \alpha\} + p(1 - \mathbb{P}\{X_n = \alpha\}) \\ &= (1 - 2p)\mathbb{P}\{X_n = \alpha\} + p < (1 - p). \end{aligned}$$

This gives,

$$\mathbb{P}(\{X_n = \alpha\} \cap \{X_{n+1} = \alpha\}) = (1 - p)\mathbb{P}\{X_n = \alpha\} > \mathbb{P}\{X_{n+1} = \alpha\}\mathbb{P}\{X_n = \alpha\},$$

showing that the events $\{X_n = \alpha\}$ and $\{X_{n+1} = \alpha\}$ are not independent. Recall that another way to express the last formula is that

$$\mathbb{P}\{X_{n+1} = \alpha | X_n = \alpha\} > \mathbb{P}\{X_{n+1} = \alpha\}.$$

Thus knowing that the strain at time n is α increases the chance that the strain at time $n+1$ is α . To study this type of process, we therefore need a new theory, which is the theory of *discrete-time Markov chains*. The possible values of X_n are making up the statespace I of the chain, here $I = \{\alpha, \beta\}$. The characteristic feature of a Markov chain is that *the past influences the future only via the present*. For example you should check yourself that

$$\mathbb{P}\{X_{n+1} = \alpha | X_n = \alpha, X_{n-1} = \alpha\} = \mathbb{P}\{X_{n+1} = \alpha | X_n = \alpha\}.$$

Here the state of the virus at time $n + 1$ (future) does not depend on the state of the virus at time $n - 1$ (past) if the state at time n (present) is already known. The question raised at the beginning of this example can be formulated as finding the value of $\mathbb{P}\{X_n = \alpha \mid X_0 = \alpha\}$. This will be computed in a problem sheet question.

Example 1.4. The simple random walk

A particle jumps about at random on the set $\mathbb{Z}$ of integers. At time 0 the particle is in fixed position $x \in \mathbb{Z}$. At each time $n \in \mathbb{N}$ a coin with probability p of heads and $q = 1 - p$ of tails is tossed. If the coin falls heads, then the particle jumps one position to the right, if the coin falls tails the particle jumps one position to the left. For $n \in \mathbb{N}$ the position X_n of the particle at time n is therefore

$$X_n = x + Y_1 + \cdots + Y_n = X_{n-1} + Y_n$$

where

$$Y_k := \begin{cases} 1 & \text{if } k\text{th toss is heads,} \\ -1 & \text{if } k\text{th toss is tails.} \end{cases}$$

The Y_k s are independent random variables with

$$\mathbb{P}\{Y_k = 1\} = p \text{ and } \mathbb{P}\{Y_k = -1\} = q.$$

The stochastic process $\{X_n : n \in \mathbb{N}\}$ has the *time-parameter set* $\mathbb{N}$ and the *statespace* is the discrete (but infinite) set $\mathbb{Z}$ of integers. The rules of evaluation are given by the laws of Y_k and we note that

$$\begin{aligned} & \mathbb{P}\{X_{n+1} = k \mid X_n = j, X_{n-1} = x_{n-1}, \dots, X_1 = x_1\} \\ &= \mathbb{P}\{Y_{n+1} = k - j \mid Y_n = j - x_{n-1}, Y_{n-1} = x_{n-1} - x_{n-2}, \dots, Y_2 = x_2 - x_1, Y_1 = x_1 - x\} \\ &= \mathbb{P}\{Y_{n+1} = k - j\} = \begin{cases} 0 & \text{if } |k - j| \neq 1, \\ p & \text{if } k - j = 1, \\ q & \text{if } k - j = -1, \end{cases} \end{aligned}$$

The right hand side is again independent of $x_1, \dots, x_{n-1}$, but depends on the value of X_n . Observe that here the independence of the random variables Y_k plays an important role, though the random variables X_n are *not* independent.

For a Markov chain $(X_n)_{n \geq 0}$, the random variables X_{n+2} and X_n are also, in general, not independent. You should check that this is the case in both examples above.

1.1.2 Specification of a discrete-time Markov chain

Suppose a chain $X = \{X_0, X_1, \dots\}$ taking values in a *discrete*, i.e. finite or countably infinite, set I . Suppose that we observe X up to time n and wish to know the distribution of a future value X_{n+1} . By (**), we know that the distribution will depend only on the value of X_n so a model for the evolution of X will require specifying *only* the transition law from any current value i_n to any future value i_{n+1} without requiring knowledge of the historical values $i_{n-1}, i_{n-2}, \dots$. This means specifying the *one-step transition probabilities* for every pair $i_n, i_{n+1} \in I$. We also need to know the initial distribution of the chain, i.e. the distribution of X_0 . Therefore, specification of a Markov chain requires the following three elements.

- (i) The *state space* I of possible values of the Markov chain at each time.

(ii) A *one-step transition matrix* $P = (p_{ij} : i, j \in I)$ with $p_{ij} \geq 0$ for all $i, j \in I$ and

$$\sum_{j \in I} p_{ij} = 1 \quad \text{for all } i \in I.$$

(iii) The *initial distribution* on the state space given by a probability mass function $w = (w_i : i \in I)$, with $w_i \geq 0$ and $\sum_{i \in I} w_i = 1$.

Theorem 1.5. *The law of evolution of the Markov chain X specified as above is given by*

$$\mathbb{P}\{X_0 = i_0, X_1 = i_1, \dots, X_n = i_n\} = w_{i_0} p_{i_0 i_1} \cdots p_{i_{n-1} i_n}, \text{ for all } i_0, \dots, i_n \in I.$$

Proof. Let $i_0, \dots, i_n \in I$. Using the multiplication formula for conditional probabilities and the Markov property,

$$\begin{aligned} \mathbb{P}\{X_0 = i_0, X_1 = i_1, \dots, X_n = i_n\} &= \mathbb{P}\{X_0 = i_0\} \times \mathbb{P}\{X_1 = i_1 \mid X_0 = i_0\} \\ &\quad \times \cdots \times \mathbb{P}\{X_{n-1} = i_{n-1} \mid X_{n-2} = i_{n-2}, \dots, X_1 = i_1, X_0 = i_0\} \\ &\quad \times \mathbb{P}\{X_n = i_n \mid X_{n-1} = i_{n-1}, \dots, X_1 = i_1, X_0 = i_0\} \\ &= \mathbb{P}\{X_0 = i_0\} \times \mathbb{P}\{X_1 = i_1 \mid X_0 = i_0\} \\ &\quad \times \cdots \times \mathbb{P}\{X_{n-1} = i_{n-1} \mid X_{n-2} = i_{n-2}\} \\ &\quad \times \mathbb{P}\{X_n = i_n \mid X_{n-1} = i_{n-1}\} \\ &= w_{i_0} p_{i_0 i_1} \cdots p_{i_{n-2} i_{n-1}} p_{i_{n-1} i_n} \end{aligned}$$

□

1.1.3 Examples

Example 1.3 cont'd. Recall the virus example. The state space is $I = \{\alpha, \beta\}$. Initially the virus is in strain α , hence $w = (w_\alpha, w_\beta) = (1, 0)$. The one-step transition matrix is

$$P = \begin{pmatrix} 1-p & p \\ p & 1-p \end{pmatrix}.$$

Example 1.4 cont'd. For the simple random walk, the state space is $I = \mathbb{Z}$. The particle starts in a fixed point $x \in \mathbb{Z}$, hence the initial distribution is given by

$$w_i = \mathbb{P}\{X_0 = i\} = \begin{cases} 1 & \text{if } i = x. \\ 0 & \text{otherwise.} \end{cases}$$

The one-step transition matrix is given by

$$p_{ij} = \mathbb{P}\{X_{n+1} = j \mid X_n = i\} = \begin{cases} 0 & \text{if } |j - i| \neq 1, \\ p & \text{if } j - i = 1, \\ q & \text{if } j - i = -1. \end{cases}$$

Example 1.6. Snakes and Ladders

Consider a game of Snakes and Ladders, where for simplicity we assume that there is a single player. The state space I is the set of squares, except those that are at the bottom of a ladder, or at the head of a snake (you immediately move from such a square). For instance, if there are six squares, with a ladder from 2 to 4 and a snake from 5 to 1, then

$$I = \{1, 3, 4, 6\}.$$

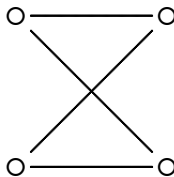
The transition matrix is determined by how many squares you move in a roll of a dice. The top row is given by:

$$P = \begin{pmatrix} 1/6 & 1/6 & 2/6 & 2/6 \\ \cdots & & & \\ \cdots & & & \\ \cdots & & & \end{pmatrix}.$$

This arises as follows. Starting at 1, if the roll is 1, 2, 3, 4, 5, 6, you move, respectively, to 4, 3, 4, 1, 6, 6. Other rows can be found similarly.

Example 1.7. Random walk on graphs

A particle is moving on the graph below by starting on the top left vertex and at each time step moving along one of the adjacent edges to a neighbouring vertex, choosing the edge with equal probability and independently of all previous movements.



There are four vertices, which we enumerate from left to right by $I = \{1, 2, 3, 4\}$. At time $n = 0$ we are in vertex 1. Hence $w = (1, 0, 0, 0)$ is the initial distribution. Each vertex has exactly two neighbours, so that it jumps to each neighbour with probability $1/2$. This gives

$$P = \begin{pmatrix} 0 & 1/2 & 0 & 1/2 \\ 1/2 & 0 & 1/2 & 0 \\ 0 & 1/2 & 0 & 1/2 \\ 1/2 & 0 & 1/2 & 0 \end{pmatrix}$$

This Markov chain is called a **simple random walk** on the graph.

One can also consider random walk on infinite graphs. The simple random walk on $\mathbb{Z}$ with $p = 1/2$ (introduced in Example 1.4) is a simple random walk on the graph whose vertex set is $\mathbb{Z}$ and where there is an edge between i and $i + 1$ for all $i \in \mathbb{Z}$. One can similarly consider simple random on $\mathbb{Z}^2$, where each vertex has four neighbours, and at each step the moves along one of the four available edges with equal probabilities. Analogous definitions apply to $\mathbb{Z}^d$, $d \geq 3$. (Indeed to any infinite graph such that every vertex has finitely many neighbours.)

Example 1.8. Card shuffling

Shuffling a deck of cards can be seen as a Markov chain. The state space I is the set of all permutations

of 52 cards, describing the position of each card in the deck, relative to a perfectly ordered deck. Thus I is huge, it has $\#I = 52!$ elements. One shuffle then corresponds to applying a permutation to the deck, that is random. One possible model for a single shuffle is the so-called **rifle shuffle**, where the deck is first cut at a random location and then the two halves are combined in a random order. In order to understand what will be a reasonable model here, imagine taking a movie of the shuffling step and playing it backwards. Then what you see is that the deck is separated into two parts at random and then one part is placed on top of the other. Let us assume that in doing this each card is placed to the left or right with equal probability. Then the size of one part will be $\text{Binom}(52, 1/2)$. Thus in the original rifle shuffle the deck should be cut at a location that is $\text{Binom}(52, 1/2)$ and the two parts combined into one with all possibilities equally likely.

Let Y be the random permutation obtained by applying a single rifle shuffle to a perfectly ordered deck. The transition matrix P can then be written as

$$p_{\pi\sigma} = \mathbb{P}[\sigma = Y \circ \pi] = \mathbb{P}[Y = \sigma \circ \pi^{-1}].$$

The initial distribution is $w_{\text{id}} = 1$ and $w_\pi = 0$ if $\pi \neq \text{id}$.

1.2 m -step transition probabilities

Unless otherwise stated $X = \{X_n : n \in \mathbb{N}\}$ is a Markov chain with (discrete) statespace I , one-step transition matrix P , and initial distribution w . Recall that

$$\mathbb{P}\{X_{n+1} = j \mid X_n = i\} = p_{ij} \quad i, j \in I, n \geq 0.$$

In order to look at multistep transitions of the Markov chain, we need the following lemma, that is a simple consequence of the Markov property (**).

Lemma 1.9. *Let B be any event depending only on $X_0, \dots, X_{n-1}$. Then*

$$\mathbb{P}\{X_{n+1} = j \mid X_n = i, B\} = p_{ij} = \mathbb{P}\{X_{n+1} = j \mid X_n = i\}.$$

Proof. Let

$$J(B) = \left\{ (j_0, \dots, j_{n-1}) : \{X_0 = j_0, \dots, X_{n-1} = j_{n-1}\} \subset B \right\}.$$

Then by the Markov property (**), we have:

$$\begin{aligned} \mathbb{P}\{X_{n+1} = j, X_n = i, B\} &= \sum_{(j_0, \dots, j_{n-1}) \in J(B)} \mathbb{P}\{X_{n+1} = j, X_n = i, X_{n-1} = j_{n-1}, \dots, X_0 = j_0\} \\ &\stackrel{(**)}{=} p_{ij} \sum_{(j_0, \dots, j_{n-1}) \in J(B)} \mathbb{P}\{X_n = i, X_{n-1} = j_{n-1}, \dots, X_0 = j_0\} \\ &= p_{ij} \mathbb{P}\{X_n = i, B\}. \end{aligned}$$

Dividing both sides by $\mathbb{P}\{X_n = i, B\}$ we obtain the claim. $\square$

Let us now consider two-step transitions:

$$\begin{aligned}
\mathbb{P}\{X_{n+2} = j \mid X_n = i\} &= \sum_{k \in I} \mathbb{P}\{X_{n+2} = j, X_{n+1} = k \mid X_n = i\} \\
&= \sum_{k \in I} \mathbb{P}\{X_{n+1} = k \mid X_n = i\} \mathbb{P}\{X_{n+2} = j \mid X_{n+1} = k, X_n = i\} \\
&\stackrel{\text{Lemma 1.9}}{=} \sum_{k \in I} p_{ik} p_{kj} = (P^2)_{ij},
\end{aligned}$$

where P^2 is the product of the matrix P with itself.

Let us also work out what happens for a three-step transition:

$$\begin{aligned}
\mathbb{P}\{X_{n+3} = j \mid X_n = i\} &= \sum_{k_1, k_2 \in I} \mathbb{P}\{X_{n+3} = j, X_{n+2} = k_2, X_{n+1} = k_1 \mid X_n = i\} \\
&= \sum_{k_1, k_2 \in I} \mathbb{P}\{X_{n+1} = k_1 \mid X_n = i\} \mathbb{P}\{X_{n+2} = k_2 \mid X_n = i, X_{n+1} = k_1\} \\
&\quad \times \mathbb{P}\{X_{n+3} = j \mid X_n = i, X_{n+1} = k_1, X_{n+2} = k_2\} \\
&= \sum_{k_1, k_2 \in I} p_{i k_1} p_{k_1 k_2} p_{k_2 j} = (P^3)_{ij},
\end{aligned}$$

More generally,

$$\boxed{\mathbb{P}\{X_{n+m} = j \mid X_n = i\} = (P^m)_{ij}.}$$

Moreover, if the vector $(w_i : i \in I)$ is the initial distribution, we get

$$\begin{aligned}
\mathbb{P}\{X_n = j\} &= \sum_{k \in I} \mathbb{P}\{X_0 = k\} \mathbb{P}\{X_n = j \mid X_0 = k\} \\
&= \sum_{k \in I} w_k (P^n)_{kj}.
\end{aligned}$$

Hence we get

$$\boxed{\mathbb{P}\{X_n = j\} = (w P^n)_j.}$$

Hence, if we can calculate the matrix power P^n , we can find the distribution of X_n and the n -step transition probabilities, i.e. the probabilities of being in state j at time $n + k$ if we are in state i at time k .

If n is large (as often we are particularly interested in the long term behaviour of the process) it is not advisable to calculate P^n directly, but there are some more efficient methods, which we shall discuss now.

1.2.1 Calculation of matrix powers by diagonalization

A matrix P is called **diagonalisable** if we can find an invertible matrix S and a diagonal matrix Λ such that

$$S^{-1} P S = \Lambda \Leftrightarrow P = S \Lambda S^{-1}.$$

Specifically, the diagonal elements $\lambda_1, \dots, \lambda_m$ of Λ are the eigenvalues of P and the columns of S are the corresponding eigenvectors. (To see this, write $\mathbf{e}_i$ for the i -th canonical basis vector, and let $\mathbf{v}_i = S\mathbf{e}_i$. Then $S^{-1}\mathbf{v}_i = \mathbf{e}_i$ and $P\mathbf{v}_i = S\Lambda S^{-1}\mathbf{v}_i = S\Lambda\mathbf{e}_i = \lambda_i S\mathbf{e}_i = \lambda_i \mathbf{v}_i$.) Not all transition matrices are diagonalisable but a sufficient condition is all eigenvalues to be distinct.

If the matrix P is diagonalisable, then

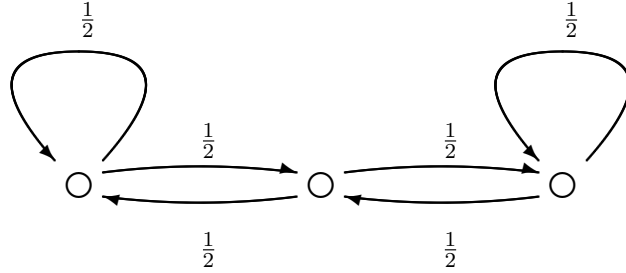
$$P^n = PP \dots P = (S\Lambda S^{-1})(S\Lambda S^{-1}) \dots (S\Lambda S^{-1}) = S\Lambda\Lambda \dots \Lambda S^{-1} = S\Lambda^n S^{-1}$$

and because $\Lambda = \text{diag}\{\lambda_1, \lambda_2, \dots, \lambda_m\}$, then $\Lambda^n = \text{diag}\{\lambda_1^n, \lambda_2^n, \dots\}$. A consequence of this is that there exist matrices $U_1, \dots, U_m$ such that

$$P^n = \lambda_1^n U_1 + \lambda_2^n U_2 + \dots + \lambda_m^n U_m.$$

(This follows by writing $\Lambda = \sum_{i=1}^m \lambda_i E_i$, where E_i is the matrix with a 1 in position (i, i) , and 0's elsewhere. Then $U_i = S E_i S^{-1}$.) It is important here that $U_1, \dots, U_m$ *do not* depend on the time n .

We present the diagonalisation method by an example. Suppose a village has three pubs along the main road. A customer decides after every pint whether he moves to the next pub on the left or on the right, and he chooses each option with the same probability. If there is no pub in this direction, he stays where he is for another pint. Here is a graph indicating the situation



The state space of this Markov chain is $I = \{1, 2, 3\}$, where the three numbers indicate the three pubs and the one-step transition matrix is

$$P = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & \frac{1}{2} & \frac{1}{2} \end{pmatrix}$$

To compute the eigenvalues of P , find $\det(\lambda I - P)$, the characteristic polynomial, of P . Here

$$\lambda I - P = \begin{pmatrix} \lambda - \frac{1}{2} & -\frac{1}{2} & 0 \\ -\frac{1}{2} & \lambda & -\frac{1}{2} \\ 0 & -\frac{1}{2} & \lambda - \frac{1}{2} \end{pmatrix},$$

and hence

$$\begin{aligned} \det(\lambda I - P) &= (\lambda - \tfrac{1}{2}) \det \begin{pmatrix} \lambda & -\frac{1}{2} \\ -\frac{1}{2} & \lambda - \frac{1}{2} \end{pmatrix} + \tfrac{1}{2} \det \begin{pmatrix} -\frac{1}{2} & 0 \\ -\frac{1}{2} & \lambda - \frac{1}{2} \end{pmatrix} \\ &= (\lambda - \tfrac{1}{2})(\lambda^2 - \lambda/2 - \tfrac{1}{4}) - \tfrac{1}{4}(\lambda - \tfrac{1}{2}) \\ &= (\lambda - \tfrac{1}{2})(\lambda - 1)(\lambda + \tfrac{1}{2}). \end{aligned}$$

Thus P has three distinct eigenvalues

$$\lambda_1 = 1, \lambda_2 = \frac{1}{2}, \lambda_3 = -\frac{1}{2},$$

so it must be diagonalisable.

Important remark: 1 is *always* an eigenvalue of P , because the vector $\mathbf{v} = (1, \dots, 1)^\top$ satisfies $P\mathbf{v} = \mathbf{v}$. This is the fact that the row sums of transition matrices are 1! Say $\lambda_1 = 1$ and so the first column of S is the vector $\mathbf{v}$. As we will show (see Lemma 1.12 to come soon) the corresponding matrix U_1 will have the form

$$U_1 = \begin{pmatrix} \pi_1 & \pi_2 & \dots \\ \pi_1 & \pi_2 & \dots \\ \vdots & \vdots & \\ \pi_1 & \pi_2 & \dots \end{pmatrix}$$

and $\pi = (\pi_1, \pi_2, \dots)$ is the first row of S^{-1} . To find π , we will use that $S^{-1}P = \Lambda S^{-1}$ so $\pi P = \pi$ and one needs to solve this equation for π . We will also have $\sum_i \pi_i = 1$. See Lemma 1.12 for the details.

We close this section with a Corollary of Lemma 1.9.

Corollary 1.10. *For any Markov chain we have*

$$\mathbb{P}\{X_{n_1} = i_1, \dots, X_{n_k} = i_k\} = (wP^{n_1})_{i_1} (P^{n_2-n_1})_{i_1 i_2} \dots (P^{n_k-n_{k-1}})_{i_{k-1} i_k}.$$

Proof. The left hand side can be written (using the multiplication rule), as

$$\begin{aligned} & \mathbb{P}\{X_{n_1} = i_1\} \mathbb{P}\{X_{n_2} = i_2 \mid X_{n_1} = i_1\} \dots \mathbb{P}\{X_{n_k} = i_k \mid X_{n_{k-1}} = i_{k-1}, \dots, X_{n_1} = i_1\} \\ &= (wP^{n_1})_{i_1} (P^{n_2-n_1})_{i_1 i_2} \dots (P^{n_k-n_{k-1}})_{i_{k-1} i_k}, \end{aligned}$$

where we used Lemma 1.9 to reduce the third, fourth, $\dots$, k -th term to transition probabilities. $\square$

Remark 1.11. Note that Definition (*) of a Markov chain also follows from Lemma 1.9.

1.2.2 Questions about the long term behaviour

- If we shuffle a deck of cards many times, the cards get “mixed up”. Mathematically, this means that all permutations become more-or-less equally likely after many shuffles. That is, we expect that $\lim_{n \rightarrow \infty} \mathbb{P}_{\text{id}}[X_n = \sigma] = \frac{1}{52!}$ for all permutations $\sigma \in I$.
- Similarly, considering a simple random walk on a finite graph, we expect that the position after a long time will “forget” where the walk started, and that there is a limiting probability to find the walker at any given vertex. Mathematically, we ask if the limit $\lim_{n \rightarrow \infty} \mathbb{P}_w[X_n = j] = \pi_j$ exists. How to find these probabilities? Is there a general statement for any Markov chain?
- What is the long term proportion of time spent in each state? Is it given by π ? Mathematically, do we have

$$\lim_{n \rightarrow \infty} \frac{1}{n} (\text{time spent in state } j \text{ during steps } 1, \dots, n) = \pi_j?$$

- How long does it take, on average, to get from some given state to another one?

Next we investigate some of these questions in simple examples, where linear algebra allows us to compute things exactly. Following this, we will look at the general theorems.

1.2.3 Tricks for Diagonalization

We continue with the same example and give some useful tricks to obtain a quick diagonalization of (small) matrices. To find the eigenvalues of the matrix P , one has three automatic equations,

- (1) 1 is always an eigenvalue for a P -matrix,
- (2) the trace of P (=sum of the diagonal entries) equals the sum of the eigenvalues,
- (3) the determinant of P is the product of the eigenvalues.

In the 2×2 or 3×3 case one may get the eigenvalues by solving the system of equations resulting from these three facts.

In our example we have $\lambda_1 = 1$ by Fact (1), Fact (2) gives $1 + \lambda_2 + \lambda_3 = 1$, which means $\lambda_2 = -\lambda_3$. Fact (3) gives $\lambda_2\lambda_3 = -1/4$, hence $\lambda_2 = 1/2$, and $\lambda_3 = -1/2$ (or vice versa).

If we have distinct eigenvalues, the matrix is diagonalizable. Then from the representation of $P^n = S\Lambda^n S^{-1}$ we see that (in the 3×3 case) there exist matrices U_1, U_2 and U_3 with

$$P^n = \lambda_1^n U_1 + \lambda_2^n U_2 + \lambda_3^n U_3.$$

We know this without having to find the eigenvectors! Now taking the values $n = 0, 1, 2$ in the equation gives

$$(1) \quad U_1 + U_2 + U_3 = P^0 = I = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

$$(2) \quad U_1 + \frac{1}{2}U_2 - \frac{1}{2}U_3 = P,$$

$$(3) \quad U_1 + \frac{1}{4}U_2 + \frac{1}{4}U_3 = P^2.$$

All one has to do is to solve the (matrix) simultaneous equations.

In the example $(1) - 4 \times (3)$ gives $-3U_1 = I - 4P^2$ and $(1) + 2 \times (2)$ gives $3U_1 + 2U_2 = I + 2P$. Now,

$$P^2 = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & \frac{1}{2} & \frac{1}{2} \end{pmatrix} \begin{pmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & \frac{1}{2} & \frac{1}{2} \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & \frac{1}{4} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ \frac{1}{4} & \frac{1}{4} & \frac{1}{2} \end{pmatrix}.$$

Hence

$$U_1 = \frac{1}{3}(4P^2 - I) = \frac{1}{3} \left[\begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{pmatrix} - \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \right] = \begin{pmatrix} \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{pmatrix}.$$

And,

$$U_2 = \frac{1}{2}(I + 2P - 3U_1) = \frac{1}{2} \left[\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} + \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix} - \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} \right] = \begin{pmatrix} \frac{1}{2} & 0 & -\frac{1}{2} \\ 0 & 0 & 0 \\ -\frac{1}{2} & 0 & \frac{1}{2} \end{pmatrix},$$

$$U_3 = I - U_1 - U_2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} - \begin{pmatrix} \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{pmatrix} - \begin{pmatrix} \frac{1}{2} & 0 & -\frac{1}{2} \\ 0 & 0 & 0 \\ -\frac{1}{2} & 0 & \frac{1}{2} \end{pmatrix} = \begin{pmatrix} \frac{1}{6} & -\frac{1}{3} & \frac{1}{6} \\ -\frac{1}{3} & \frac{2}{3} & -\frac{1}{3} \\ \frac{1}{6} & -\frac{1}{3} & \frac{1}{6} \end{pmatrix}.$$

Hence, for all $n = 0, 1, 2, \dots$,

$$P^n = \begin{pmatrix} \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{pmatrix} + \left(\frac{1}{2}\right)^n \begin{pmatrix} \frac{1}{2} & 0 & -\frac{1}{2} \\ 0 & 0 & 0 \\ -\frac{1}{2} & 0 & \frac{1}{2} \end{pmatrix} + \left(-\frac{1}{2}\right)^n \begin{pmatrix} \frac{1}{6} & -\frac{1}{3} & \frac{1}{6} \\ -\frac{1}{3} & \frac{2}{3} & -\frac{1}{3} \\ \frac{1}{6} & -\frac{1}{3} & \frac{1}{6} \end{pmatrix}.$$

And recall that the n -step transition probabilities are just the entries of this matrix, i.e.

$$p_{ij}^{(n)} := \mathbb{P}\{X_n = j \mid X_0 = i\} = (P^n)_{ij},$$

for example $\mathbb{P}\{X_n = 2 \mid X_0 = 0\} = (P^n)_{02} = \frac{1}{3} - \frac{1}{2} \left(\frac{1}{2}\right)^n + \frac{1}{6} \left(-\frac{1}{2}\right)^n$.

As a further example look at

$$P = \begin{pmatrix} 0 & 1 & 0 \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ 0 & 1 & 0 \end{pmatrix}.$$

Then the eigenvalues are $\lambda_1 = 1$ (always) and λ_2, λ_3 satisfy $\lambda_1 \lambda_2 \lambda_3 = \det P = 0$, and $\lambda_1 + \lambda_2 + \lambda_3 = \text{trace } P = 1/2$. Hence $\lambda_2 = -1/2$ and $\lambda_3 = 0$. As the eigenvalues are distinct, the matrix P is diagonalizable and there exist matrices U_1, U_2, U_3 with

$$P^n = U_1 + \left(-\frac{1}{2}\right)^n U_2 + 0^n U_3, \text{ for all } n = 0, 1, 2, \dots$$

Note: We define $0^0 = 1$.

Either take $n = 0, 1, 2$ and solve or find U_1 directly as

$$U_1 = \begin{pmatrix} \frac{1}{6} & \frac{2}{3} & \frac{1}{6} \\ \frac{1}{6} & \frac{2}{3} & \frac{1}{6} \\ \frac{1}{6} & \frac{2}{3} & \frac{1}{6} \end{pmatrix},$$

by solving $\pi P = \pi$. For the latter, see “Important remark” in Section 1.2.1, and also the general lemma further below. In the second case one only has to use the equations for $n = 0, 1$ and gets $I = U_1 + U_2 + U_3$ and $P = U_1 - \frac{1}{2}U_2$. Hence

$$U_2 = 2(U_1 - P) = \begin{pmatrix} \frac{1}{3} & -\frac{2}{3} & \frac{1}{3} \\ -\frac{1}{6} & \frac{1}{3} & -\frac{1}{6} \\ \frac{1}{3} & -\frac{2}{3} & \frac{1}{3} \end{pmatrix}.$$

For $n \geq 1$ we do not need to find U_3 and get

$$P^n = \begin{pmatrix} \frac{1}{6} & \frac{2}{3} & \frac{1}{6} \\ \frac{1}{6} & \frac{2}{3} & \frac{1}{6} \\ \frac{1}{6} & \frac{2}{3} & \frac{1}{6} \end{pmatrix} + \left(-\frac{1}{2}\right)^n \begin{pmatrix} \frac{1}{3} & -\frac{2}{3} & \frac{1}{3} \\ -\frac{1}{6} & \frac{1}{3} & -\frac{1}{6} \\ \frac{1}{3} & -\frac{2}{3} & \frac{1}{3} \end{pmatrix}.$$

Let us explore some of the consequences of diagonalization. Since

$$P^n = U_1 + \lambda_2^n U_2 + \dots + \lambda_m^n U_m,$$

if we also have $|\lambda_2|, \dots, |\lambda_m| < 1$, then $P^n \rightarrow U_1$ as $n \rightarrow \infty$. This applies in both examples above and provides us with the long term behaviour: $\lim_{n \rightarrow \infty} \mathbb{P}_i[X_n = j] = \lim_{n \rightarrow \infty} (P^n)_{ij} = (U_1)_{ij}$. How to find U_1 easily?

Lemma 1.12. Assume that P is diagonalisable and $|\lambda_2|, \dots, |\lambda_m| < 1$. There exists a unique row vector π such that $\pi P = \pi$ and $\sum_{i \in I} \pi_i = 1$. Also,

$$U_1 = \begin{pmatrix} \pi_1 & \dots & \pi_m \\ \vdots & & \vdots \\ \pi_1 & \dots & \pi_m \end{pmatrix}.$$

Proof. As noted earlier, the columns of S are the right eigenvectors of P . Choose S in such a way that its first column is the vector with all entries equal to 1 (which is an eigenvector corresponding to the eigenvalue $\lambda_1 = 1$). Let π equal the first row of S^{-1} . Then since $S^{-1}P = \Lambda S^{-1}$ (where $\Lambda = \text{diag}(1, \lambda_2, \dots, \lambda_m)$), we have $\pi P = \lambda_1 \pi = 1 \cdot \pi = \pi$. We also have

$$\sum_{i=1}^m \pi_i = (\text{top row of } S^{-1}) \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix} = (S^{-1}S)_{11} = I_{11} = 1.$$

From this

$$\begin{aligned} U_1 &= SE_1S^{-1} = \begin{pmatrix} 1 & & \\ \vdots & & \\ 1 & & \end{pmatrix} \begin{pmatrix} 1 & & 0 \\ \vdots & \ddots & \vdots \\ 0 & & 0 \end{pmatrix} \begin{pmatrix} \pi_1 & \dots & \pi_m \\ & & \\ & & \end{pmatrix} \\ &= \begin{pmatrix} 1 & & \\ \vdots & & \\ 1 & & \end{pmatrix} \begin{pmatrix} \pi_1 & \dots & \pi_m \\ \vdots & & \vdots \\ 0 & \dots & 0 \end{pmatrix} = \begin{pmatrix} \pi_1 & \dots & \pi_m \\ \vdots & & \vdots \\ \pi_1 & \dots & \pi_m \end{pmatrix}. \end{aligned}$$

□

Corollary 1.13. When P is diagonalisable and $|\lambda_2|, \dots, |\lambda_m| < 1$, then for all $i, j \in I$ we have

$$\mathbf{P}[X_n = j \mid X_0 = i] = (P^n)_{ij} \xrightarrow{n \rightarrow \infty} \pi_j,$$

where π solves $\pi P = \pi$ and $\sum_{i \in I} \pi_i = 1$. The vector π has non-negative entries.

Questions:

- (i) Under what conditions on the Markov chain can we conclude that the limit $\lim_{n \rightarrow \infty} \mathbb{P}_i[X_n = j] = \pi_j$ exists? Can we understand this convergence probabilistically?
- (ii) Does this also apply when I is countably infinite?

(The rest of this subsection is non-examinable)

We close this section by discussing what happens when not all eigenvalues are distinct. A **Jordan block** with eigenvalue λ is a matrix of the form

$$J = \begin{pmatrix} \lambda & 1 & & & \\ & \lambda & 1 & & \\ & & \ddots & \ddots & \\ & & & \lambda & 1 \\ & & & & \lambda \end{pmatrix}$$

Theorem (Jordan canonical form) *Given any matrix A , there exists an invertible matrix S such that $A = S L S^{-1}$, where L is a block-diagonal matrix with Jordan blocks. Each eigenvalue of A occurs as many times on the diagonal of L in total as its algebraic multiplicity, i.e. its multiplicity as a root of the characteristic polynomial $\chi_A(\lambda) := \det(\lambda I - A)$.*

Proof. Non-examinable — See Linear Algebra Texts. □

The Jordan canonical form implies that we can write P^n in a form:

$$P^n = U_1 + \cdots + \lambda_i^n U_i + \binom{n}{1} \lambda_i^{n-1} U_i^{(1)} + \cdots$$

Lemma 1.14. (a) *If J is a Jordan block with eigenvalue λ , then*

$$J^n = \begin{pmatrix} \lambda^n & n\lambda^{n-1} & \binom{n}{2}\lambda^{n-2} & \cdots & \cdots \\ & \lambda^n & n\lambda^{n-1} & \binom{n}{2}\lambda^{n-2} & \cdots \\ & & \ddots & \ddots & \ddots \\ & & & \lambda^n & n\lambda^{n-1} & \binom{n}{2}\lambda^{n-2} \\ & & & & \lambda^n & n\lambda^{n-1} \\ & & & & & \lambda^n \end{pmatrix}$$

In other words,

$$(J^n)_{ij} = \begin{cases} \binom{n}{k} \lambda^{n-k} & \text{if } j = i + k, k \geq 0; \\ 0 & \text{otherwise.} \end{cases}$$

(b) *If all eigenvalues of A have absolute value smaller than 1, then $\lim_{n \rightarrow \infty} A^n = 0$ (elementwise), and $(I - A)^{-1}$ exists.*

Proof. (a) Left as an exercise.

(b) From the Jordan canonical form: $A^n = S L^n S^{-1}$, where L^n is block-diagonal, with n -th powers of Jordan blocks. In each block, if $|\lambda| < 1$, we have $|\binom{n}{k} \lambda^{n-k}| \rightarrow 0$ as $n \rightarrow \infty$, hence $A^n \rightarrow 0$. Since 1 is not an eigenvalue, $(I - A)^{-1}$ exists. □

1.2.4 The method of generating functions

This method is particularly efficient when we want to find certain entries of the matrix P^n without having to determine the complete matrix. The starting point is the geometric series,

$$1 + z + z^2 + \cdots = \frac{1}{1 - z}, \text{ for all } |z| < 1.$$

We have the following analogue for matrices:

Fact. *If A is a square matrix such that $\lim_{n \rightarrow \infty} A^n = 0$ (elementwise), and $I - A$ is invertible, then*

$$I + A + A^2 + \cdots = (I - A)^{-1}.$$

Proof. The proof mimicks the geometric series argument. Write:

$$(I - A)(I + A + \cdots + A^N) = (I + A + \cdots + A^N) - (A + A^2 + \cdots + A^{N+1}) = I - A^{N+1},$$

which gives

$$\sum_{k=0}^n A^k = (I - A)^{-1}(I - A^{N+1}).$$

Letting $N \rightarrow \infty$ yields the formula. □

Recall that the one-step transition matrix of a finite state Markov chain does *not* fulfill either condition, as 1 is always an eigenvalue. However, we have the following useful fact.

Lemma 1.15. *Suppose P is the one-step transition matrix of a finite state Markov chain, then all eigenvalues have modulus less or equal to one.*

Proof. See Sheet 2. □

We can thus use the geometric series for the matrices θP for all $|\theta| < 1$. The idea is to expand $(I - \theta P)^{-1}$ as a power series in θ around $\theta = 0$ and to find P^n by comparing the coefficients of θ^n .

Let us look at an example. Let X be a Markov chain with statespace $I = \{0, 1, 2\}$ and one-step transition matrix

$$P = \begin{pmatrix} 0 & 1 & 0 \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ 0 & 1 & 0 \end{pmatrix}.$$

Then

$$I - \theta P = \begin{pmatrix} 1 & -\theta & 0 \\ -\frac{1}{4}\theta & 1 - \frac{1}{2}\theta & -\frac{1}{4}\theta \\ 0 & -\theta & 1 \end{pmatrix}.$$

To invert this matrix, recall that $A^{-1} = C^T / \det A$ where $C = \{c_{ij} : 0 \leq i, j \leq 2\}$ is the matrix of cofactors of A , i.e. c_{ij} is the determinant of the matrix A with the i th row and j th column removed, multiplied with the factor $(-1)^{i+j}$. For example,

$$\begin{aligned} (I - \theta P)_{01}^{-1} &= \frac{c_{10}}{\det(I - \theta P)} = \frac{-\det \begin{pmatrix} -\theta & 0 \\ -\theta & 1 \end{pmatrix}}{\det \begin{pmatrix} 1 - \frac{1}{2}\theta & -\frac{1}{4}\theta \\ -\theta & 1 \end{pmatrix} - (-\theta) \det \begin{pmatrix} -\frac{1}{4}\theta & -\frac{1}{4}\theta \\ 0 & 1 \end{pmatrix}} \\ &= \frac{\theta}{1 - \frac{1}{2}\theta - \frac{1}{4}\theta^2 - \frac{1}{4}\theta^2} = \frac{\theta}{(1 - \theta)(1 + \frac{1}{2}\theta)}. \end{aligned}$$

To expand this as a power series in θ we need to use partial fractions, write

$$\frac{\theta}{(1 - \theta)(1 + \frac{1}{2}\theta)} = \frac{a}{1 - \theta} + \frac{b}{1 + \frac{1}{2}\theta}.$$

then $\theta = a(1 + \frac{1}{2}\theta) + b(1 - \theta)$. Using this for $\theta = 1$ gives $a = 2/3$, and for $\theta = 0$ we get $b = -2/3$. We can thus continue with

$$(I - \theta P)_{01}^{-1} = \frac{2}{3}(1 - \theta)^{-1} - \frac{2}{3}(1 - (-\theta/2))^{-1} = \frac{2}{3}(1 + \theta + \theta^2 + \dots) - \frac{2}{3}(1 - \frac{\theta}{2} + \frac{\theta^2}{4} - \frac{\theta^3}{8} + \dots).$$

For the last expansion we have used the geometric series. Now we can compare the coefficients here with those coming from the matrix valued geometric series,

$$(I - \theta P)_{01}^{-1} = \sum_{n=0}^{\infty} \theta^n (P^n)_{01} = \sum_{n=0}^{\infty} \theta^n p_{01}^{(n)},$$

and get, for all $n \in \mathbb{N}$,

$$\mathbb{P}\{X_n = 1 \mid X_0 = 0\} = p_{01}^{(n)} = \frac{2}{3} - \frac{2}{3} \left(-\frac{1}{2}\right)^n.$$

You may wish to check that $p_{01}^{(0)} = 0$.

As another example we ask for the probability that, starting in state 0, we are again in state 0 after n time steps. Then,

$$(I - \theta P)_{00}^{-1} = \frac{c_{00}}{\det(I - \theta P)} = \frac{\det \begin{pmatrix} 1 - \frac{\theta}{2} & -\frac{\theta}{4} \\ -\theta & 1 \end{pmatrix}}{(1 - \theta)(1 + \frac{\theta}{2})} = \frac{1 - \frac{\theta}{2} - \frac{\theta^2}{4}}{(1 - \theta)(1 + \frac{\theta}{2})}.$$

Now we use partial fractions again (note that we need the α -term, as numerator and denominator have the same order),

$$\frac{1 - \frac{\theta}{2} - \frac{\theta^2}{4}}{(1 - \theta)(1 + \frac{\theta}{2})} = \alpha + \frac{\beta}{1 - \theta} + \frac{\gamma}{1 + \frac{\theta}{2}}.$$

To get α, β, γ , we look at

$$\alpha(1 - \theta)(1 + \frac{\theta}{2}) + \beta(1 + \frac{\theta}{2}) + \gamma(1 - \theta) = 1 - \frac{\theta}{2} - \frac{\theta^2}{4}.$$

Then $\theta = 1$ gives $\beta = 1/6$, $\theta = -2$ gives $\gamma = 1/3$ and comparing the θ^2 coefficient yields $\alpha = 1/2$. Hence,

$$\begin{aligned} (I - \theta P)_{00}^{-1} &= \frac{1}{2} + \frac{1}{6}(1 - \theta)^{-1} + \frac{1}{3} \left(1 - \left(-\frac{\theta}{2}\right)\right)^{-1} \\ &= \frac{1}{2} + \frac{1}{6}(1 + \theta + \theta^2 + \theta^3 + \dots) + \frac{1}{3} \left(1 - \frac{\theta}{2} + \frac{\theta^2}{4} - \dots\right). \end{aligned}$$

Equating coefficients of θ^n gives,

$$\mathbb{P}\{X_n = 0 \mid X_0 = 0\} = \begin{cases} \frac{1}{6} + \frac{1}{3} \left(-\frac{1}{2}\right)^n & \text{for } n \geq 1, \\ 1 & \text{for } n = 0. \end{cases}$$

1.3 Hitting probabilities and expected waiting times

As in the previous sections, X is a Markov chain with discrete statespace I and one-step transition matrix P . We now introduce a notation which allows us to look at the Markov chain and vary the initial distributions, this point of view turns out to be very useful later.

We denote by $\mathbb{P}_w$ the law of the chain X when the initial distribution is w . In other words, for all $i_0, i_1, \dots, i_n \in I$,

$$\mathbb{P}_w\{X_0 = i_0, X_1 = i_1, \dots, X_n = i_n\} = w_{i_0} p_{i_0 i_1} p_{i_1 i_2} \dots p_{i_{n-1} i_n}.$$

We denote by $\mathbb{P}_i$ the law of the chain X when X starts in the fixed state $i \in I$. Thus for all $i_0, i_1, \dots, i_n \in I$,

$$\mathbb{P}_i\{X_0 = i_0, X_1 = i_1, \dots, X_n = i_n\} = \begin{cases} p_{i i_1} p_{i_1 i_2} \cdots p_{i_{n-1} i_n} & \text{when } i_0 = i; \\ 0 & \text{when } i_0 \neq i. \end{cases}$$

We can also think of $\mathbb{P}_i$ as the law of the Markov chain conditioned on $X_0 = i$, i.e.

$$\mathbb{P}_i(A) = \mathbb{P}\{A \mid X_0 = i\}.$$

The law $\mathbb{P}_w$ of X with initial distribution w is then given as the *mixture* or weighted average of the laws $\mathbb{P}_i$, i.e. for the initial distribution $w = (w_i : i \in I)$ of the Markov chain and any event A ,

$$\mathbb{P}_w(A) = \sum_{i \in I} \mathbb{P}_w(A \cap \{X_0 = i\}) = \sum_{i \in I} \mathbb{P}_w\{X_0 = i\} \mathbb{P}_i(A) = \sum_{i \in I} w_i \mathbb{P}_i(A).$$

We also use this notation for expected values, i.e. $\mathbb{E}_i$ refers to the expectation with respect to $\mathbb{P}_i$ and $\mathbb{E}_w$ refers to expectation with respect to $\mathbb{P}_w$.

Remember that in expressions such as $\mathbb{P}\{X_{n+1} = x_{n+1} \mid X_n = x_n\} = p_{x_n x_{n+1}}$ we do not have to write the index w because if we are told the state at time n we can forget about the initial distribution.

Example. For states i, j, k we have

$$\begin{aligned} \mathbb{P}[X_7 = k, X_5 = j \mid X_2 = i] &= \mathbb{P}[X_5 = j \mid X_2 = i] \mathbb{P}[X_7 = k \mid X_5 = j] = \mathbb{P}_i[X_3 = j] \mathbb{P}_j[X_2 = k] \\ &= p_{ij}^{(3)} p_{jk}^{(2)}. \end{aligned}$$

1.3.1 Hitting probabilities

We define the first hitting time T_j of a state $j \in I$ by

$$T_j := \inf\{n \geq 1 : X_n = j\} \in \{1, 2, \dots\} \cup \{\infty\},$$

and understand that $T_j := \infty$ if the set of n 's is empty. Note that we exclude time 0: we always have $T_j > 0$. Also observe the following equality of events,

$$\{T_j = \infty\} = \{X \text{ never hits } j\}.$$

Using this we can now define the *hitting probabilities* $(F_{ij} : i, j \in I)$, where F_{ij} is the probability that the chain started in i hits the state j (in positive, finite time). We let

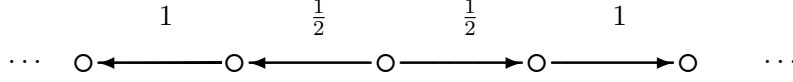
$$F_{ij} := \mathbb{P}_i\{T_j < \infty\}.$$

Note that F_{ii} is the probability that the chain started in i returns to i in finite time. This need not be 1, as the following simple example shows.

Example 1.16. We suppose a particle is moving on the integers $\mathbb{Z}$ as follows: it is started at the origin and in the first step moves to the left with probability $1/2$ and to the right with probability $1/2$. In each further step it moves one step away from the origin. (Exercise: write down the P -matrix!)

Here we have

$$F_{ij} = \begin{cases} 1 & \text{if } j > i > 0 \text{ or } j < i < 0, \\ 0 & \text{if } i < 0 < j \text{ or } i > 0 > j, \\ 1/2 & \text{if } i = 0, j \neq 0, \\ 0 & \text{if } i = j. \end{cases}$$



Theorem 1.17. For a fixed state $b \in I$ the probabilities

$$x_i := F_{i,b} := \mathbb{P}_i\{T_b < \infty\}, \text{ for } i \in I,$$

form the minimal non-negative solution of the system of equations,

$$x_i = \left(\sum_{j \neq b} p_{ij} x_j \right) + p_{ib} \quad \text{for } i \in I. \quad (1.3.1)$$

In particular, if $y_i \geq 0$ for all $i \in I$ form a solution of

$$y_i = \left(\sum_{j \neq b} p_{ij} y_j \right) + p_{ib}, \quad (1.3.2)$$

then $y_i \geq x_i$ for every $i \in I$.

Note: $y_i = 1$ for all $i \in I$ always solves (1.3.2).

Proof. First step: We show that $x_i = \mathbb{P}_i\{T_b < \infty\}$ is a solution of (1.3.1). Let $E = \{T_b < \infty\}$ be the event that the chain hits b . Then

$$\begin{aligned}
 x_i &= \mathbb{P}_i(E) = \sum_{j \in I} \mathbb{P}_i(E \cap \{X_1 = j\}) \\
 &= \sum_{j \in I} \mathbb{P}_i\{X_1 = j\} \mathbb{P}_i\{E \mid X_1 = j\} \\
 &= \sum_{j \in I} p_{ij} \mathbb{P}\{E \mid X_1 = j\}.
 \end{aligned}$$

Looking at the last probability we see that

$$\mathbb{P}\{E \mid X_1 = j\} = \begin{cases} 1 & \text{if } j = b, \\ \mathbb{P}_j\{T_b < \infty\} & \text{if } j \neq b. \end{cases}$$

This gives

$$x_i = \sum_{j \neq b} p_{ij} x_j + p_{ib}.$$

Second step: Suppose $y_i \geq 0$ forms a solution of (1.3.2). To finish the proof we have to show that $y_i \geq x_i$. Observe that,

$$\begin{aligned}
 y_i &= \left(\sum_{j \neq b} p_{ij} y_j \right) + p_{ib} \\
 &= p_{ib} + \sum_{j \neq b} p_{ij} \left(\sum_{k \neq b} p_{jk} y_k + p_{jb} \right) \\
 &= p_{ib} + \sum_{j \neq b} p_{ij} p_{jb} + \sum_{j \neq b} \sum_{k \neq b} p_{ij} p_{jk} y_k \\
 &= \mathbb{P}_i\{X_1 = b\} + \mathbb{P}_i\{X_1 \neq b, X_2 = b\} + \sum_{j \neq b} \sum_{k \neq b} p_{ij} p_{jk} y_k.
 \end{aligned}$$

Repeated substitution yields,

$$\begin{aligned}
 y_i &= \mathbb{P}_i\{X_1 = b\} + \mathbb{P}_i\{X_1 \neq b, X_2 = b\} + \cdots + \mathbb{P}_i\{X_1 \neq b, \dots, X_{n-1} \neq b, X_n = b\} \\
 &\quad + \sum_{j_1 \neq b} \cdots \sum_{j_n \neq b} p_{ij_1} p_{j_1 j_2} \cdots p_{j_{n-1} j_n} y_{j_n}.
 \end{aligned}$$

Because $y_j \geq 0$ the last term is non-negative and we get, for all $n \in \mathbb{N}$,

$$\begin{aligned}
 y_i &\geq \mathbb{P}_i\{X_1 = b\} + \mathbb{P}_i\{X_1 \neq b, X_2 = b\} + \cdots + \mathbb{P}_i\{X_1 \neq b, \dots, X_{n-1} \neq b, X_n = b\} \\
 &= \mathbb{P}_i\{T_b = 1\} + \cdots + \mathbb{P}_i\{T_b = n\} \\
 &= \mathbb{P}_i\{T_b \leq n\}.
 \end{aligned}$$

Hence, letting $n \rightarrow \infty$,

$$y_i \geq \lim_{n \rightarrow \infty} \mathbb{P}_i\{T_b \leq n\} = \mathbb{P}_i\left(\bigcup_n \{T_b \leq n\}\right) = \mathbb{P}_i\{T_b < \infty\} = x_i.$$

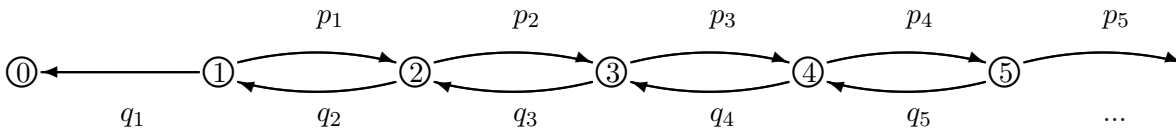
□

Example 1.18. ('One-dimensional' Markov chains)

Consider a Markov chain with state space $I = \{0, 1, 2, \dots\}$ given by the diagram below, where, for $i = 1, 2, \dots$, we have $0 < p_i = 1 - q_i < 1$. Here 0 is an absorbing state and

$$x_i := \mathbb{P}_i\{T_0 < \infty\}$$

is the hitting probability of state 0 if the chain is started in state $i \in I$.



We characterize the hitting probabilities in terms of the following quantities. Let

$$\gamma_0 := 1, \gamma_i := \frac{q_i q_{i-1} \cdots q_1}{p_i p_{i-1} \cdots p_1} \text{ for } i \geq 1.$$

$$\gamma_1 = \frac{q_1}{p_1}$$

- If $\sum_{i=0}^{\infty} \gamma_i = \infty$ we have $x_i = 1$ for all $i \in I$.
- If $\sum_{i=0}^{\infty} \gamma_i < \infty$, we have

$$x_i = \frac{\sum_{j=i}^{\infty} \gamma_j}{\sum_{j=0}^{\infty} \gamma_j} \text{ for } i \in I.$$

To prove this, we first use the ‘one-step method’ to find the equations $x_0 = 1$ and

$$x_i = q_i x_{i-1} + p_i x_{i+1}, \quad \text{for } i \geq 1.$$

We show that the solutions of this system are given by

$$x_i = 1 - A(\gamma_0 + \cdots + \gamma_{i-1}), \quad \text{for } i \geq 1,$$

for any choice of a fixed $A \in \mathbb{R}$. To prove this we have to

- check that the given form solves the equation,
- show that any solution is of this form.

For the first part we just plug in:

$$\begin{aligned} q_i x_{i-1} + p_i x_{i+1} &= q_i (1 - A(\gamma_0 + \cdots + \gamma_{i-2})) + p_i (1 - A(\gamma_0 + \cdots + \gamma_i)) \\ &= 1 - A(q_i(\gamma_0 + \cdots + \gamma_{i-2}) + p_i(\gamma_0 + \cdots + \gamma_i)), \end{aligned}$$

and

$$q_i(\gamma_0 + \cdots + \gamma_{i-2}) + p_i(\gamma_0 + \cdots + \gamma_i) = \gamma_0 + \cdots + \gamma_{i-2} + \frac{p_i q_{i-1} \cdots q_1 + q_i \cdots q_1}{p_{i-1} \cdots p_1} = \gamma_0 + \cdots + \gamma_{i-1}.$$

For the second part assume that $(x_i : i = 1, 2, \dots)$ is *any* solution. Let $y_i = x_{i-1} - x_i$. Then $p_i y_{i+1} = q_i y_i$, which implies inductively that

$$y_{i+1} = \left(\frac{q_i}{p_i}\right) y_i = \gamma_i y_1.$$

Hence

$$x_0 - x_i = y_1 + \cdots + y_i = y_1(\gamma_0 + \cdots + \gamma_{i-1}).$$

Therefore, for $A := y_1$, we have

$$x_i = 1 - A(\gamma_0 + \cdots + \gamma_{i-1}).$$

To complete the proof we have to find the *smallest* nonnegative solution, which means we want to make A as large as possible without making x_i negative. First suppose that $\sum_{i=0}^{\infty} \gamma_i = \infty$, then for any $A > 0$ the solution gets negative eventually, so that the largest nonnegative solution corresponds to the case $A = 0$, i.e. $x_i = 1$.

Next suppose that $\sum_{i=0}^{\infty} \gamma_i := M < \infty$. Then the solution remains nonnegative iff $1 - AM \geq 0$, i.e. if $A \leq 1/M$, so that the choice $A = 1/M$ gives the smallest solution. This solution is the one giving the right value for the hitting probabilities.

$$1 - \frac{1}{n}(\gamma_0 + \dots + \gamma_{i-1}) = 1 - \frac{1}{\sum_{j=0}^{\infty} \gamma_j}(\gamma_0 + \dots + \gamma_{i-1}) = \frac{2 - 2}{2}$$

Example 1.19. Now we look at a simple random walk $\{X_n : n = 0, 1, 2, \dots\}$ with parameter $p \in [1/2, 1]$. If $X_0 = i > 0$, we are essentially in the situation of the previous example with $p_i = p$ and $q_i = 1 - p$. Then $\gamma_i = ((1-p)/p)^i$ and $\sum_{i=0}^{\infty} \gamma_i$ diverges if $p = \frac{1}{2}$ and converges otherwise. Hence $F_{i0} = 1$ if $p = \frac{1}{2}$ and otherwise, for $i > 0$,

$$F_{i0} = \frac{\sum_{j=i}^{\infty} \left(\frac{1-p}{p}\right)^j}{\sum_{j=0}^{\infty} \left(\frac{1-p}{p}\right)^j} = \left(\frac{1-p}{p}\right)^i.$$

Note that, by the one-step method,

$$F_{00} = p F_{10} + q F_{-10} = p \frac{1-p}{p} + q = 1 - p + q = 2 - 2p.$$

Hence we can summarize our results for the random walk case as follows.

Theorem 1.20. The hitting probabilities $F_{ij} = \mathbb{P}_i\{T_j < \infty\}$ for a simple random walk with parameter $p \in [1/2, 1]$ are given in the symmetric case $p = 1/2$ by

$$F_{ij} = 1 \quad \text{for all } i, j \in \mathbb{Z},$$

and in the asymmetric case $p > 1/2$ by

$$F_{ij} = \begin{cases} 1 & \text{if } i < j, \\ 2 - 2p & \text{if } i = j, \\ \left(\frac{1-p}{p}\right)^{i-j} & \text{if } i > j. \end{cases}$$

By looking at $-X$ the theorem also gives full information about the case $p < 1/2$. As a special case note that, for all $p \in [0, 1]$,

$$\mathbb{P}_0\{X \text{ does not return to } 0\} = |p - q|.$$

Example 1.21. (A queue with two servers) Customers line up in a single queue for two available servers. Consider the number of customers in the queue, including any customers being served. Assume the following transition mechanism for changes in the queue length, where $\mu > 0$ is a parameter.

If the queue is currently empty, one person arrives next with probability 1;

If the queue has one customer, $\begin{cases} \text{a new person arrives with probability } 1/(1+\mu); \\ \text{the customer finishes with probability } \mu/(1+\mu); \end{cases}$

If the queue has at least two customers, $\begin{cases} \text{a new person arrives with probability } 1/(1+2\mu); \\ \text{one customer finishes with probability } 2\mu/(1+2\mu). \end{cases}$

Here we have $p_1 = \frac{1}{1+\mu}$, $q_1 = \frac{\mu}{1+\mu}$, and $p_i = \frac{1}{1+2\mu}$, $q_i = \frac{2\mu}{1+2\mu}$, $i \geq 2$. Therefore,

$$\sum_{i=0}^{\infty} \gamma_i = 1 + \mu + \mu(2\mu) + \mu(2\mu)^2 + \dots = \begin{cases} 1 + \frac{\mu}{1-2\mu} < \infty & \text{when } \mu < \frac{1}{2}; \\ \infty & \text{when } \mu \geq \frac{1}{2}. \end{cases}$$

Thus

$$\begin{aligned} F_{i0} &= \mathbb{P}[\text{with } i \text{ customers initially, the queue becomes empty eventually}] \\ &= \begin{cases} 1 & \text{when } \mu \geq \frac{1}{2}; \\ < 1 & \text{when } \mu < \frac{1}{2}. \end{cases} \end{aligned}$$

1.3.2 Expected hitting times

In a manner similar to the problem of hitting probabilities, Theorem 1.17, one can prove the following theorem for the expected time until a state b is hit for the first time.

Theorem 1.22. *Fix a state $b \in I$ and let $y_i := \mathbb{E}_i\{T_b\}$. Then*

$$y_i = 1 + \sum_{j \neq b} p_{ij} y_j \quad (1.3.3)$$

and y is the minimal nonnegative solution of this equation.

Note that $y_i = \infty$ is possible for some, or even all, i , and the convention $0 \times \infty = 0$ is in place.

Proof. Let us first see that $\mathbb{E}_i\{T_b\}$ solve (1.3.3). Write $T_b = 1 + \tilde{T}_b$, where 1 stands for the first step of the walk (recall that in the definition of T_b we require the time to be ≥ 1), and $\tilde{T}_b$ stands for the remaining time. When the first step is to b , then $\tilde{T}_b = 0$, and when it is to $j \neq b$, it is the number of steps taken from j to hit b . Thus using the Markov property at time 1, we have

$$\begin{aligned} \mathbb{E}_i\{T_b\} &= 1 + \mathbb{E}_i\{\tilde{T}_b\} = 1 + \sum_{j \neq b} \mathbb{E}_i\{\mathbf{1}_{X_1=j} \tilde{T}_b\} = 1 + \sum_{j \neq b} \mathbb{P}_i\{X_1 = j\} \mathbb{E}_i\{\tilde{T}_b \mid X_1 = j\} \\ &= 1 + \sum_{j \neq b} p_{ij} \mathbb{E}_j\{T_b\}. \end{aligned}$$

Now let us prove minimality. Repeated substitution into the equation (1.3.3) gives:

$$\begin{aligned} y_i &= 1 + \sum_{i_1 \neq b} p_{i i_1} y_{i_1} \\ &= 1 + \sum_{i_1 \neq b} p_{i i_1} + \sum_{i_1, i_2 \neq b} p_{i i_1} p_{i_1 i_2} y_{i_2} \\ &= 1 + \sum_{i_1 \neq b} p_{i i_1} + \sum_{i_1, i_2 \neq b} p_{i i_1} p_{i_1 i_2} y_{i_2} + \sum_{i_1, i_2, i_3 \neq b} p_{i i_1} p_{i_1 i_2} p_{i_2 i_3} y_{i_3} \\ &\vdots \\ &= 1 + \sum_{i_1 \neq b} p_{i i_1} + \sum_{i_1, i_2 \neq b} p_{i i_1} p_{i_1 i_2} y_{i_2} + \cdots + \sum_{i_1, \dots, i_{n-1} \neq b} p_{i i_1} p_{i_1 i_2} \cdots p_{i_{n-2} i_{n-1}} \\ &\quad + \sum_{i_1, \dots, i_n \neq b} p_{i i_1} p_{i_1 i_2} \cdots p_{i_{n-1} i_n} y_{i_n}. \end{aligned}$$

Since $y_{i_n} \geq 0$, the last term is non-negative. This gives:

$$y_i \geq \mathbb{P}_i[T_b \geq 1] + \mathbb{P}_i[T_b \geq 2] + \cdots + \mathbb{P}_i[T_b \geq n].$$

Letting $n \rightarrow \infty$ we get $y_i \geq \sum_{m=1}^{\infty} \mathbb{P}_i[T_b \geq m] = \mathbb{E}_i[T_b]$. This proves minimality. $\square$

Example 1.23. Consider a Markov chain with statespace $I = \{A, B, C, D\}$ and jump probabilities given by the diagram in Figure 1.1

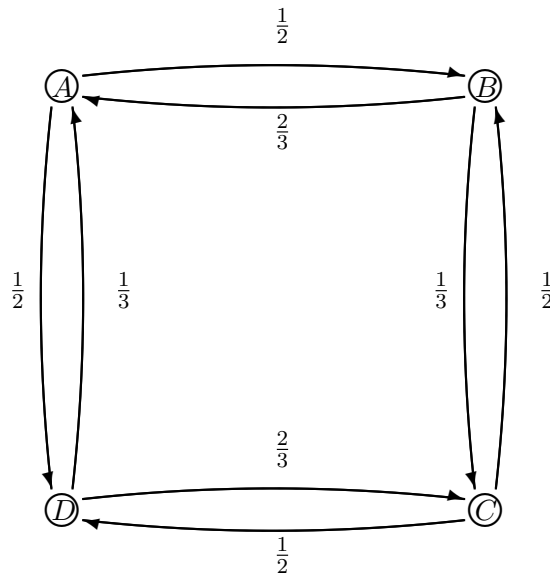


Figure 1.1: Diagram for Example 1.23.

Problem 1: Find the expected time until the chain started in C reaches A .

Solution: Let $x_i = \mathbb{E}_i\{T_A\}$. By considering the first step and using the Markov property (or just using (1.3.3)) we get

$$\begin{aligned} x_C &= 1 + \frac{1}{2}x_B + \frac{1}{2}x_D \\ x_B &= 1 + \frac{2}{3}0 + \frac{1}{3}x_C \\ x_D &= 1 + \frac{1}{3}0 + \frac{2}{3}x_C. \end{aligned}$$

Hence,

$$x_C = 1 + \frac{1}{2}(1 + \frac{1}{3}x_C) + \frac{1}{2}(1 + \frac{2}{3}x_C) = 2 + \frac{1}{2}x_C,$$

which implies $\mathbb{E}_C\{T_A\} = x_C = 4$. The expected time until the chain started in C reaches A is 4.

Problem 2: What is the probability that the chain started in A reaches the state C before B ?

Solution: Let $x_i = \mathbb{P}_i\{T_C < T_B\}$. By considering the first step and using the Markov property we get

$$\begin{aligned} x_A &= \frac{1}{2}x_D + \frac{1}{2}0, \\ x_D &= \frac{1}{3}x_A + \frac{2}{3}1. \end{aligned}$$

Hence $2x_A = (1/3)x_A + (2/3)$, which gives $x_A = 2/5$. Hence the probability of hitting C before B when we start in A is $2/5$.

For some further examples, we need to look at the following question:

Question: Does the Markov property generalize to *random* times? For example, we might expect that

$$\mathbb{P}_i[\text{return to } i, \text{ and then return a second time}] = \mathbb{P}_i[T_i < \infty] \mathbb{P}_i[T_i < \infty],$$

and this is indeed always true, as we will see. However, the same does not work with arbitrary random times, as the following simple counterexample shows. Let $T_i^{\text{last}} := \sup\{n \geq 0 : X_n = i\}$ be the *last* time i is visited. Then, if there is positive probability for a last return to i at a finite time, then

$$\mathbb{P}_i[\text{visit } i \text{ after } T_i^{\text{last}}] = 0 \neq \mathbb{P}_i[T_i^{\text{last}} < \infty] \mathbb{P}_i[T_i < \infty].$$

What random times are OK? This is clarified by the following notion.

Definition 1.24. We write $\mathcal{F}_n$ = events determined by $X_0, \dots, X_n$. A random variable $T : \Omega \rightarrow \{0, 1, 2, \dots\} \cup \{\infty\}$ is called a **stopping time** for the Markov chain, if for all $n \geq 0$ we have $\{T = n\} \in \mathcal{F}_n$.

Thus a stopping time is a random time that ‘does not look into the future’: we can decide when it happens based only on the observed values of the Markov chain up to that time. For example:

$$\{T_i = 2\} = \{X_1 \neq i, X_2 = i\} = \cup_{i_0, i_1 \in I} \{X_0 = i_0, X_1 = i_1, X_2 = i\} \in \mathcal{F}_2.$$

More generally, we have the following.

Example 1.25. The first hitting time T_i is a stopping time, because

$$\{T_i = n\} = \{X_1 \neq i, \dots, X_{n-1} \neq i, X_n = i\} \in \mathcal{F}_n.$$

On the other hand, in general, T_i^{last} is *not* a stopping time, because

$$\{T_i^{\text{last}} = n\} = \{X_n = i, X_{n+1} \neq i, X_{n+2} \neq i, \dots\},$$

and thus the values of $X_{n+1}, X_{n+2}, \dots$ are typically needed.

Theorem 1.26. (Strong Markov Property) *Let T be a stopping time for the Markov chain $(X_n)_{n \geq 0}$. Conditional on $T < \infty$ and $X_T = i$, the law of $X_T, X_{T+1}, \dots$ is given by $\mathbb{P}_i$, and is independent of $X_0, \dots, X_{T-1}$.*

Proof. (Non-examinable) We compute

$$\begin{aligned} & \mathbb{P}[T < \infty, X_T = i, X_{T+1} = i_1, \dots, X_{T+n} = i_n] \\ &= \sum_{m=0}^{\infty} \mathbb{P}[T = m, X_m = i, X_{m+1} = i_1, \dots, X_{m+n} = i_n]. \end{aligned}$$

Observe that the events $T = m$ and $X_m = i$ only involve $X_0, \dots, X_m$ (using that T is a stopping time). Thus we can use the Markov property at time m to get that the right hand side equals

$$\sum_{m=0}^{\infty} \mathbb{P}[T = m, X_m = i] \mathbb{P}_i[X_1 = i_1, \dots, X_n = i_n].$$

This proves the claim about the law of $X_T, X_{T+1}, \dots$. A similar computation shows independence: for any $j_0, \dots, j_{m-1} \in I$ we have:

$$\begin{aligned} & \mathbb{P}[T = m, X_0 = j_0, \dots, X_{m-1} = j_{m-1}, X_m = i, X_{m+1} = i_1, \dots, X_{m+n} = i_n] \\ &= \mathbb{P}[T = m, X_0 = j_0, \dots, X_{m-1} = j_{m-1}] \mathbb{P}_i[X_1 = i_1, \dots, X_n = i_n]. \end{aligned}$$

□

Here are some examples using the Strong Markov Property.

Example 1.27. (a) Consider a simple random walk on $\mathbb{Z}$, which steps to the right with probability p . If $p > 1/2$, we found earlier that $\mathbb{P}_1[T_0 < \infty] = \frac{1-p}{p}$. The Strong Markov Property implies that for $i \geq 2$ we have

$$\mathbb{P}_i[T_0 < \infty] = \mathbb{P}_i[T_{i-1} < \infty] \mathbb{P}_{i-1}[T_0 < \infty].$$

Thus by induction on i , we get $\mathbb{P}_i[T_0 < \infty] = (\frac{1-p}{p})^i$.

(b) Similarly, for $i \geq 1$ we have

$$\mathbb{E}_i[T_0] = \mathbb{E}_i[T_{i-1}] + \mathbb{E}_{i-1}[T_0] = \mathbb{E}_1[T_0] + \mathbb{E}_{i-1}[T_0] \cdots = i \mathbb{E}_1[T_0].$$

(c) *Important to note:* If $p = 1/2$, we have

$$\mathbb{P}_0[T_0 < \infty] = 1 = \sum_{n=1}^{\infty} \mathbb{P}_0[T_0 = n],$$

but

$$\mathbb{E}_0[T_0] = \infty = \sum_{n=1}^{\infty} n \mathbb{P}_0[T_0 = n].$$

To see the second claim, write out the first step of the walk:

$$\mathbb{E}_0[T_0] = 1 + \frac{1}{2} \mathbb{E}_1[T_0] + \frac{1}{2} \mathbb{E}_{-1}[T_0] = 1 + \mathbb{E}_1[T_0]$$

by symmetry. Putting $y_i = \mathbb{E}_i[T_0]$, we have the equation:

$$y_1 = 1 + \frac{1}{2} y_2 = 1 + \frac{1}{2} 2 y_1 = 1 + y_1.$$

This has no finite non-negative solution, hence $y_1 = \mathbb{E}_1[T_0] = \infty$. It follows that $\mathbb{E}_0[T_0] = 1 + y_1 = \infty$. More generally, for $p = 1/2$ we get:

The average waiting time until a symmetric, simple random walk travels from state i to state j is infinite for all $i \neq j$.

However, we know that the waiting time is finite almost surely!

(d) If $p < 1/2$ (the case of a downward drift) we have

$$y_1 = 1 + p y_2 = 1 + 2p y_1.$$

We get the solution $y_1 = 1/(1 - 2p)$, and hence

$$\mathbb{E}_n\{T_0\} = \frac{n}{1-2p} \text{ for all } n \geq 1.$$

Example 1.28. Consider a simple random walk on the infinite graph constructed in the following way. We start with a square $OCAB$. To each vertex we attach a new square, and to each three remaining vertices of those square we attach a further square. Continue attaching squares indefinitely. The question is to find $\mathbb{P}_A[T_0 < \infty]$.

Solution: Write $a = \mathbb{P}_A[T_0 < \infty]$, $b = \mathbb{P}_B[T_0 < \infty] = \mathbb{P}_C[T_0 < \infty]$ (the last equality due to symmetry of the graph. For any vertex $Y \neq O$, we have $\mathbb{P}_Y[T_0 < \infty] = a^{n_Y} b^{m_Y}$ for some integers $n_Y, m_Y \geq 0$. This is because for every $Y \neq O$, there is a unique sequence of squares the walk must cross in order to hit O , and the Strong Markov Property implies that we pick up a factor a when a single edge of the square must be crossed, and a factor b , when the square must be crossed diagonally. Using this property, we get the equations:

$$\begin{aligned} a &= \frac{1}{4}b + \frac{1}{4}b + \frac{1}{4}ba + \frac{1}{4}ba = \frac{1}{2}b + \frac{1}{2}ba = \frac{1}{2}b(1 + a) \\ b &= \frac{1}{4} + \frac{1}{4}a + \frac{1}{4}b^2 + \frac{1}{4}b^2 = \frac{1}{4} + \frac{1}{4}a + \frac{1}{2}b^2. \end{aligned}$$

Eliminating a yields the cubic equation:

$$0 = b^3 - 4b^2 + 4b - 1 = (b - 1)(b^2 - 3b + 1).$$

The minimal non-negative solution is then given by $a = \frac{3-\sqrt{5}}{1+\sqrt{5}}$, $b = \frac{3-\sqrt{5}}{2}$.

1.4 Classification of states and convergence to equilibrium

We now begin our study of the long-term behaviour of the Markov chain. Recall that $T_i = \inf\{n > 0 : X_n = i\}$ and

$$F_{ii} = \mathbb{P}_i\{T_i < \infty\} = \mathbb{P}_i\{X \text{ returns to } i \text{ at some positive time}\}.$$

Definition The state i is called

- *transient* if $\mathbb{P}_i[T_i < \infty] < 1$, i.e. if there is a positive probability of escape from i ,
- *recurrent* or *persistent* if $\mathbb{P}_i[T_i < \infty] = 1$, i.e. if the chain returns to state i almost surely.

Note: The Strong Markov Property gives:

$$\mathbb{P}_i[\text{at least } k \text{ returns to } i] = \underbrace{\mathbb{P}_i[T_i < \infty] \cdots \mathbb{P}_i[T_i < \infty]}_{k \text{ times}} = F_{ii}^k, \quad k \geq 1.$$

Hence, if i is recurrent, we have

$$\mathbb{P}_i[\text{return to } i \text{ infinitely often}] = 1,$$

and when i is transient, we have

$$\mathbb{P}_i[\text{return to } i \text{ exactly } k \text{ times}] = F_{ii}^k(1 - F_{ii}), \quad k \geq 0.$$

Let us look at the *total number of visits* to a state i given by the random variable

$$V_i = \sum_{n=0}^{\infty} \mathbf{1}_{\{X_n=i\}},$$

where $\mathbf{1}_{\{X_n=i\}}$ takes the value 1 if $X_n = i$ and 0 otherwise. Note that we include time $n = 0$.

If i is recurrent we have $\mathbb{P}_i\{V_i = \infty\} = 1$, in particular the expected number of visits to state i is $\mathbb{E}_i\{V_i\} = \infty$. If i is transient, then, for $n \geq 1$,

$$\mathbb{P}_i\{V_i = n\} = F_{ii}^{n-1}(1 - F_{ii}),$$

hence

$$\mathbb{E}_i\{V_i\} = \sum_{n=1}^{\infty} n\mathbb{P}_i\{V_i = n\} = (1 - F_{ii}) \sum_{n=1}^{\infty} nF_{ii}^{n-1} = \frac{1 - F_{ii}}{(1 - F_{ii})^2} = \frac{1}{1 - F_{ii}} < \infty,$$

recalling $\sum_{n=1}^{\infty} nx^{n-1} = (1 - x)^{-2}$ for $|x| < 1$.

1.4.1 Communicating classes

Definition 1.29. (1) We say that **state j can be reached from i** , denoted $i \longrightarrow j$, if there exists $n \geq 0$ such that $p_{ij}^{(n)} > 0$.

(2) We say that **i and j communicate**, denoted $i \longleftrightarrow j$, if both $i \longrightarrow j$ and $j \longrightarrow i$.

Lemma 1.30. *The relation $\longleftrightarrow$ is an equivalence relation on the state space I .*

Proof. • For all $i \in I$ we have $i \longleftrightarrow i$, since $p_{ii}^{(0)} = 1 > 0$.

- If $i \longleftrightarrow j$, then also $j \longleftrightarrow i$ (since the definition of $\longleftrightarrow$ is symmetric in i and j).
- If $i \longleftrightarrow j$ and $j \longleftrightarrow k$, then we have $p_{ij}^{(n)} > 0$ and $p_{jk}^{(m)} > 0$ for some $n, m \geq 0$. Hence

$$\begin{aligned} (P^{n+m})_{ik} &= \mathbb{P}_i\{X_{n+m} = k\} \\ &\geq \mathbb{P}_i\{X_n = j, X_{n+m} = k\} \\ &= \mathbb{P}_i\{X_n = j\} \mathbb{P}_i\{X_{n+m} = k \mid X_n = j\} \\ &= (P^n)_{ij} (P^m)_{jk} > 0. \end{aligned}$$

Similarly, there exist n', m' with $(P^{n'+m'})_{ki} > 0$ and hence $i \longleftrightarrow k$. □

Since $\leftrightarrow$ is an equivalence relation, we can consider the corresponding equivalence classes, which are called the **communicating classes** of the transition matrix P . The class of i consists of all $j \in I$ with $i \longleftrightarrow j$. A property of a state is called a **class property** if whenever it holds for one state, it holds for all states in the same class.

Theorem 1.31. *Recurrence and transience are class properties, i.e., in each communicating class, either all states are recurrent, or all states are transient.*

Proof. (i) *Recurrence is a class property.* Let $i \in I$ be a recurrent state. We show that if $i \longrightarrow j$, then j is also recurrent and $F_{ij} = F_{ji} = 1$. (This is a slightly stronger statement than showing that $i \longleftrightarrow j$ implies that j is recurrent.) We have

$$\mathbb{P}_i[T_j < T_i < \infty] > 0.$$

Indeed, if this probability was 0, we had

$$\mathbb{P}_i[j \text{ is hit before } k\text{-th return to } i] = 0, \quad k \geq 1,$$

(by the Strong Markov Property), and then

$$\mathbb{P}_i[T_j < \infty] = \lim_{k \rightarrow \infty} \mathbb{P}_i[j \text{ is hit before } k\text{-th return to } i] = 0.$$

This would contradict $i \longrightarrow j$. By independence of parts of the path in between returns to i , we have

$$\mathbb{P}_i[j \text{ is hit before } k\text{-th return to } i] = 1 - (1 - \mathbb{P}_i[T_j < T_i < \infty])^k \xrightarrow{k \rightarrow \infty} 1,$$

and hence $F_{ij} = 1$. If we had $F_{ji} < 1$, then we would get

$$1 - F_{ii} = \mathbb{P}_i[T_i = \infty] \geq \mathbb{P}_i[T_j < T_i] \mathbb{P}_j[T_i = \infty] > 0,$$

contradicting recurrence of i . Thus $F_{ji} = 1$ as well. It follows then that

$$F_{jj} \geq \mathbb{P}_j[T_i < \infty] \mathbb{P}_i[T_j < \infty] = F_{ji} F_{ij} = 1,$$

and j is recurrent.

(ii) *Transience is a class property.* If i is transient and $j \longleftrightarrow i$, then j cannot be recurrent, due to part (i). Thus j is transient. □

Definition 1.32. A Markov chain is called *irreducible* if all states communicate with each other, i.e. if I is the only communicating class.

Thanks to the decomposition, the study of general chains is frequently reduced to the study of irreducible chains. We can decompose the statespace I as a disjoint union

$$I = T \cup R_1 \cup R_2 \cup \dots,$$

where T is the *set* of transient states and $R_1, R_2, \dots$ are the recurrent classes.

If X starts in T , it can either stay in T forever (somehow drifting off to infinity) or get trapped (and stay forever) in one of the recurrent classes.

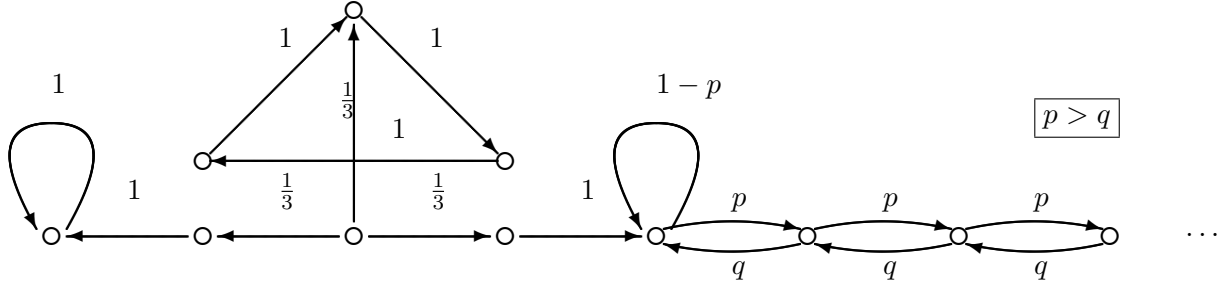


Figure 1.2: 6 classes, 2 recurrent, 4 transient

1.4.2 Period

Definition 1.33. The **period** of state $i \in I$ is defined as

$$d(i) := \text{g.c.d.} \left\{ n \geq 1 : p_{ii}^{(n)} > 0 \right\},$$

where ‘g.c.d.’ stands for ‘greatest common divisor’. The state i is called **aperiodic**, if $d(i) = 1$.

Example 1.34. Simple random walk on $\mathbb{Z}^d$. All states have period 2: call $v \in \mathbb{Z}^d$ *even*, if $\sum_{r=1}^d v_r$ is even, and call it *odd* otherwise. Starting at an even/odd vertex, the random vertex will always be at an even/odd vertex (respectively) at all even times.

Example 1.35. For the following transition matrix, all states have period 3. Positive entries are marked by a *.

$$P = \begin{pmatrix} 0 & 0 & * & * & 0 & 0 \\ 0 & 0 & * & * & 0 & 0 \\ 0 & 0 & 0 & 0 & * & * \\ 0 & 0 & 0 & 0 & * & * \\ * & * & 0 & 0 & 0 & 0 \\ * & * & 0 & 0 & 0 & 0 \end{pmatrix}$$

Example 1.36. Simple random walk on the vertices of a triangle is aperiodic: $p_{ii}^{(n)} > 0$ when $n = 2, 3, 4, \dots$

Lemma 1.37. For every $i \in I$ there exists $n_0 = n_0(i) \geq 1$ such that

$$p_{ii}^{(n d(i))} > 0 \quad \text{for all } n \geq n_0.$$

Proof. (Non-examinable) First assume $d(i) = 1$. Let $S = \{n \geq 1 : p_{ii}^{(n)} > 0\}$. Then due to properties of the greatest common divisor (see e.g. the Euclidean algorithm), there exist $\ell \geq 1$ and $n_1, \dots, n_\ell \in S$ and $c_1, \dots, c_\ell \in \mathbb{Z}$ such that

$$1 = \text{g.c.d.} S = c_1 n_1 + \dots + c_\ell n_\ell.$$

Let $c := \max\{|c_s| : 1 \leq s \leq \ell\}$. Put $N = c n_1 + \dots + c n_\ell$. We claim that $N, N+1 \in S$. Indeed: on the one hand

$$p_{ii}^{(N)} \geq \left(p_{ii}^{(n_1)}\right)^c \cdots \left(p_{ii}^{(n_\ell)}\right)^c > 0.$$

On the other hand, $N + 1 = (c + c_1)n_1 + \dots + (c + c_\ell)n_\ell$, so

$$p_{ii}^{(N+1)} \geq \left(p_{ii}^{(n_1)}\right)^{c+c_1} \dots \left(p_{ii}^{(n_\ell)}\right)^{c+c_\ell} > 0.$$

Put $n_0 = N^2$. Then any $n \geq n_0$ can be written as

$$n = N(m + N - k) + k(N + 1) \quad \text{for some integers } m \geq 0 \text{ and } 0 \leq k < N.$$

Here $m + N - k \geq 0$, $k \geq 0$, so $n \in S$ follows from $N, N + 1 \in S$. This proves the Lemma when $d(i) = 1$.

When $d(i) \geq 2$, we can divide all elements of S by $d(i)$, and apply the same argument. $\square$

Lemma 1.38. *If P is irreducible, then each $i \in I$ has the same period.*

Proof. Fix $i, j \in I$ so that $p_{ij}^{(n)} > 0$ and $p_{ji}^{(m)} > 0$ for some $n, m \geq 1$. Then

$$p_{ii}^{(n+m)} \geq p_{ij}^{(n)} p_{ji}^{(m)} > 0,$$

and hence $d(i) | (n + m)$. Let $n_0 = n_0(i)$ be as in Lemma 1.37. Then if $N \geq n_0$, we have

$$p_{jj}^{(n+m+Nd(i))} \geq p_{ji}^{(m)} p_{ii}^{(Nd(i))} p_{ij}^{(n)} > 0.$$

Here $d(i) | (n + m + Nd(i))$, and all multiples of $d(i)$ greater than $n + m + n_0 d(i)$ appear. Hence $d(j) | d(i)$. Interchanging the roles of i and j we get $d(i) | d(j)$, and therefore $d(i) = d(j)$. $\square$

1.5 Invariant distributions and convergence to equilibrium

In this section P will be an irreducible transition matrix on I . Due to Lemma 1.38, then all states have the same period d . When $d = 1$, we call P aperiodic.

Definition 1.39. A non-negative vector $\eta = (\eta_i : i \in I)$ is call **invariant**, if $\eta P = \eta$, where $(\eta P)_i = \sum_{j \in I} \eta_j p_{ji}$.

We will write $\eta \geq 0$ to indicate that each component of η is non-negative, that is, $\eta_i \geq 0$, $i \in I$.

We will be interested in finding (when it exists) an invariant *distribution* $\pi = (\pi_i : i \in I)$, that is, in addition to $\pi P = \pi$, $\pi \geq 0$ we also have $\sum_{i \in I} \pi_i = 1$. The main goal of this section will be to clarify the conditions under which π exists, and to describe the large n behaviour of $p_{ij}^{(n)}$ in terms of π .

1.5.1 Positive Recurrence

Given $i \in I$, we define

$$\begin{aligned} \gamma_j &= \gamma_j^{(i)} := \mathbb{E}_i \left[\sum_{n=0}^{\infty} \mathbf{1}_{X_n=j, n < T_i} \right] \\ &= \mathbb{E}_i[\# \text{ visits to } j \text{ before } T_i]. \end{aligned}$$

Proposition 1.40. *Let P be an irreducible transition matrix on I .*

(i) *If $\eta P = \eta$, $\eta \geq 0$ and $\eta_i = 1$, then $\eta \geq \gamma^{(i)}$.*

(ii) *If in addition i is recurrent, then $\eta = \gamma^{(i)}$ (and hence η is unique).*

Proof. (Non-examinable)

(i) For any $j \neq i$ We have

$$\begin{aligned} \eta_j &= (\eta P)_j = \eta_i p_{ij} + \sum_{j_1 \neq i} \eta_{j_1} p_{j_1 j} \\ &= p_{ij} + \sum_{j_1 \neq i} \eta_i p_{i j_1} p_{j_1 j} + \sum_{j_1, j_2 \neq i} \eta_{j_2} p_{j_2 j_1} p_{j_1 j} \\ &\vdots \\ &= p_{ij} + \sum_{j_1 \neq i} \eta_i p_{i j_1} p_{j_1 j} + \cdots + \sum_{j_1, \dots, j_{n-1} \neq i} p_{i j_{n-1}} \cdots p_{j_1 j} + \sum_{j_1, \dots, j_n \neq i} \eta_{j_n} p_{j_n j_{n-1}} \cdots p_{j_1 j}. \end{aligned}$$

Since the last term is non-negative, we get

$$\begin{aligned} \eta_j &\geq \mathbb{P}_i[X_1 = j, 1 < T_i] + \mathbb{P}_i[X_2 = j, 2 < T_i] + \cdots + \mathbb{P}_i[X_n = j, n < T_i] \\ &= \sum_{t=1}^n \mathbb{P}_i[X_t = j, t < T_i] \\ &= \sum_{t=0}^n \mathbb{P}_i[X_t = j, t < T_i]. \end{aligned}$$

The last equality is because $X_0 = i \neq j$, so the $t = 0$ term is 0. Letting $n \rightarrow \infty$ we get

$$\eta_j \geq \sum_{t=0}^{\infty} \mathbb{P}_i[X_t = j, t < T_i] = \mathbb{E}_i \left[\sum_{t=0}^{\infty} \mathbf{1}_{X_t=j, t < T_i} \right] = \gamma_j.$$

(ii) Suppose i is recurrent. Then for all $j \in I$ we have

$$\gamma_j = \mathbb{E}_i \left[\sum_{t=1}^{\infty} \mathbf{1}_{X_t=j, t \leq T_i} \right].$$

This is because we can swap the $t = 0$ term for the $t = T_i$ term, since $T_i < \infty$ with probability 1. Then the right hand side further equals

$$\sum_{k \in I} \mathbb{E}_i \left[\sum_{t=1}^{\infty} \mathbf{1}_{X_{t-1}=k, X_t=j, t-1 < T_i} \right].$$

Here the event $\{X_{t-1} = k, t-1 < T_i\}$ only involves $X_0, \dots, X_{t-1}$. Thus by the Markov property at time $t-1$ we get that the above expression equals

$$\sum_{k \in I} \mathbb{E}_i \left[\sum_{t=1}^{\infty} \mathbf{1}_{X_{t-1}=k, t-1 < T_i} \right] p_{kj} = \sum_{k \in I} \mathbb{E}_i \left[\sum_{t=0}^{\infty} \mathbf{1}_{X_t=k, t < T_i} \right] p_{kj} = (\gamma P)_j.$$

The calculation above shows that $\gamma = \gamma P$. Thus letting $\delta = \eta - \gamma$, we have $\delta \geq 0$ (by part (i)), and $\delta_i = 0$. This gives

$$0 = \delta_i = (\delta P)_i = (\delta P^2)_i = \cdots = (\delta P^n)_i.$$

Hence for any $n \geq 1$, $0 = \sum_k \delta_k p_{ki}^{(n)}$, so $\delta_k = 0$ for any $k \in I$ such that $p_{ki}^{(n)} > 0$. Due to irreducibility, this is all of the state space, so $\delta \equiv 0$ and $\eta = \gamma$.

□

Note that

$$\sum_{j \in I} \gamma_j^{(i)} = \mathbb{E}_i[\# \text{ visits to all of } I \text{ before } T_i] = \mathbb{E}_i[T_i] =: m_i.$$

Definition 1.41. Let $i \in I$ be recurrent. We call i **positive recurrent**, if $\mathbb{E}_i T_i < \infty$. We call i **null recurrent**, if $\mathbb{E}_i T_i = \infty$.

Theorem 1.42. Let P be irreducible on I . The following are equivalent:

- (i) every state is positive recurrent;
- (ii) some state $i \in I$ is positive recurrent;
- (iii) there exists an invariant distribution π .

When these hold, we have $\pi_i = \frac{1}{\mathbb{E}_i T_i} = \frac{1}{m_i}$ for all $i \in I$.

Proof.

(i) $\Rightarrow$ (ii): This is trivial.

(ii) $\Rightarrow$ (iii): Put $\pi_j := \gamma_j^{(i)} / \sum_{k \in I} \gamma_k^{(i)}$, which is a distribution. Then by Proposition 1.40(ii) we have $\gamma P = \gamma$. This implies $\pi P = \pi$. Note that we also have $\pi_i = \gamma_i^{(i)} / m_i = 1/m_i$.

(iii) $\Rightarrow$ (i): Given π , let $i \in I$ be such that $\pi_i > 0$. Put $\eta_j = \pi_j / \pi_i$. Then $\pi P = \pi$ implies $\eta P = \eta$, and hence by Proposition 1.40(i) we get

$$m_i = \sum_{k \in I} \gamma_k^{(i)} \leq \sum_{k \in I} \frac{\pi_k}{\pi_i} = \frac{1}{\pi_i} < \infty.$$

Hence i is positive recurrent, and in particular recurrent. Then from Proposition 1.40(ii) we have $\eta_j = \frac{\pi_j}{\pi_i} = \gamma_j^{(i)}$, so $\pi_j = \pi_i \gamma_j^{(i)} > 0$ for all $j \in I$. Thus we can repeat the argument with j in place of i and get that any state j is positive recurrent.

□

1.5.2 Convergence to equilibrium

If π is an invariant distribution, we have $\pi = \pi P = \pi P^2 = \dots = \pi P^n$ for every $n \geq 1$. Thus

$$\mathbb{P}_\pi[X_n = j] = (\pi P^n)_j = \pi_j.$$

For this reason, π is also called a **stationary distribution** for the Markov chain: X_n has distribution π for all $n \geq 0$.

Theorem 1.43 (Convergence to equilibrium). *Let P be irreducible on I and aperiodic. Suppose that P is positive recurrent. Then for any initial distribution $w = (w_i : i \in I)$ we have*

$$\lim_{n \rightarrow \infty} \mathbb{P}_w[X_n = j] = \pi_j = \frac{1}{m_j}, \quad j \in I,$$

where π is the invariant distribution. In particular,

$$\lim_{n \rightarrow \infty} p_{ii}^{(n)} = \pi_i = \frac{1}{\mathbb{E}_i T_i}, \quad i \in I.$$

Due to the above theorem, π is also called the **equilibrium distribution**, referring to the fact that the chain ‘reaches equilibrium’ starting from any initial distribution.

Proof. Let $(Y_n)_{n \geq 0}$ be a Markov chain with the same transition matrix P , but with initial distribution π , run independently of $(X_n)_{n \geq 0}$.

Idea: Coupling: after some time it will occur for some n that $X_n = Y_n$. After this, X_n should behave ‘just as’ Y_n .

The pair $(X_n, Y_n)_{n \geq 0}$ is a Markov chain with state space $\bar{I} := I \times I$, and transition matrix $\bar{P}$ given by

$$\bar{p}((i, j), (k, \ell)) = p_{ik} p_{j\ell}, \quad i, j, k, \ell \in I.$$

Irreducibility and aperiodicity of P implies that $\bar{P}$ is irreducible (see Problem Sheets).

We claim that $\bar{P}$ is positive recurrent. This follows simply from the fact that $\bar{\pi}(i, j) := \pi_i \pi_j$ is invariant for $\bar{P}$:

$$\sum_{(i, j) \in \bar{I}} \pi_i \pi_j p_{ik} p_{j\ell} = \pi_k \pi_\ell,$$

so $\bar{\pi} \bar{P} = \bar{\pi}$.

Set $\bar{T} := \inf\{n \geq 0 : X_n = Y_n\}$, and let

$$Z_n := \begin{cases} X_n & \text{if } n < \bar{T}; \\ Y_n & \text{if } n \geq \bar{T}. \end{cases}$$

Then $(Z_n)_{n \geq 0}$ also follows the transition matrix P , with initial distribution w . We have

$$\mathbb{P}[Z_n = j] = \mathbb{P}[X_n = j \text{ and } n < \bar{T}] + \mathbb{P}[Y_n = j \text{ and } n \geq \bar{T}].$$

Therefore, we have

$$\begin{aligned} |\mathbb{P}_w[X_n = j] - \pi_j| &= |\mathbb{P}[Z_n = j] - \mathbb{P}[Y_n = j]| \\ &= |\mathbb{P}[X_n = j \text{ and } n < \bar{T}] - \mathbb{P}[Y_n = j \text{ and } n < \bar{T}]| \\ &\leq \mathbb{P}[\bar{T} > n]. \end{aligned} \tag{1.5.1}$$

In the last step we used that we have a difference of the probabilities of two events that are both contained in the event $\{\bar{T} > n\}$.

If we can show that $\mathbb{P}[\bar{T} < \infty] = 1$, we are done, because then the right hand side of (1.5.1) goes to 0, as $n \rightarrow \infty$. Observe that we have

$$\bar{T} \leq \tilde{T}_{(i_0, i_0)} := \inf\{n \geq 0 : (X_n, Y_n) = (i_0, i_0)\}.$$

Therefore, it is enough to show that $\mathbb{P}[\tilde{T}_{(i_0, i_0)} < \infty] = 1$. Since $\bar{P}$ is recurrent, we have

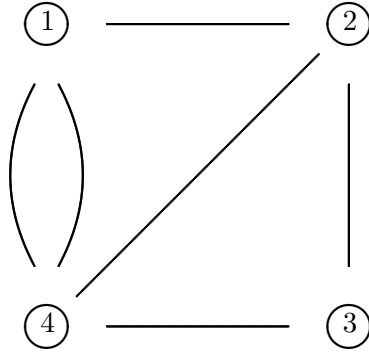
$$\mathbb{P}_{(i, j)}[\tilde{T}_{(i_0, i_0)} < \infty] = 1 \quad \text{for all } (i, j) \in \bar{I}.$$

Thus

$$\mathbb{P}[\tilde{T}_{(i_0, i_0)} < \infty] = \sum_{(i, j) \in \bar{I}} w_i \pi_j \mathbb{P}_{(i, j)}[\tilde{T}_{(i_0, i_0)} < \infty] = 1,$$

as required. □

Example 1.44. Consider a simple random walk on the graph shown in the figure below:



In the next section we are going to prove that on any finite graph $G = (V, E)$ without loop-edges, we have $\pi_v = \frac{\deg(v)}{2|E|}$, where $\deg(v) = \#$ edges incident with v . Thus in the present example, we immediately obtain: $\pi = (\frac{3}{12}, \frac{3}{12}, \frac{2}{12}, \frac{4}{12})$. Irreducibility is clear, since the graph is connected. Aperiodicity is also clear: it is possible to return to vertex 1, say, in 2 and 3 steps. The walk is positive recurrent (since π exists). Thus Theorem 1.43 gives that for all $i = 1, 2, 3, 4$ we have

$$\lim_{n \rightarrow \infty} \mathbb{P}_i[X_n = j] = \pi_j = \begin{cases} 3/12 & \text{when } j = 1; \\ 3/12 & \text{when } j = 2; \\ 2/12 & \text{when } j = 3; \\ 4/12 & \text{when } j = 4. \end{cases}$$

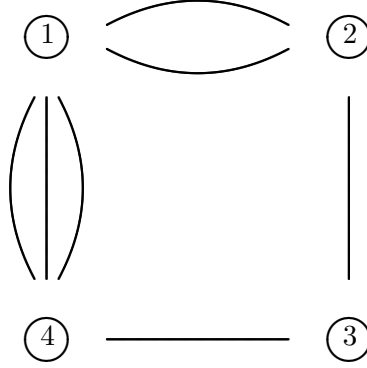
How does the convergence theorem change, if the period is $d \geq 2$?

Lemma 1.45. *If P is irreducible with period d , then P^d has d communicating classes $C_0, \dots, C_{d-1}$, with the property that if $i \in C_s$, $j \in C_t$ with $t - s \equiv r \pmod{d}$, then $p_{ij}^{(md+r')} = 0$, unless $r' = r$ (here $0 \leq r, r' < d$).*

Proof. (Non examinable)

Sketch: For any $n, m \geq 0$ such that $p_{ij}^{(n)} > 0$ and $p_{ji}^{(m)} > 0$ we have $p_{ii}^{(n+m)} > 0$, which gives $d|(n+m)$ and hence $m \equiv -n \pmod{d}$. Thus for any $i, j \in I$ there exists a remainder $0 \leq r(i, j) < d$ such that $p_{ij}^{(n)} = 0$ unless $n \equiv r \pmod{d}$. Fix some $i_0 \in I$, and put $C_r = \{j \in I : r(i_0, j) = r\}$. Concatenating paths we see that the claim holds with these sets. $\square$

Example 1.46. Consider a simple random walk on the graph shown in the figure below:



Irreducibility is clear, since the graph is connected. The period is $d = 2$. We have $C_0 = \{1, 3\}$ and $C_1 = \{2, 4\}$. $p_{11}^{(2m+1)} = 0$, $p_{13}^{(2m+1)} = 0$ for all $m \geq 0$. We also have $p_{12}^{(2m)} = 0$ and $p_{14}^{(2m)} = 0$ for all $m \geq 0$.

Theorem 1.47. *Let P be irreducible with period d , and positive recurrent. If $i \in C_s$, $j \in C_t$ with $t - s \equiv r \pmod{d}$, $0 \leq r < d$, then*

$$\lim_{m \rightarrow \infty} p_{ij}^{(md+r)} = d\pi_j.$$

Proof. Let

$$T_k^{(d)} = \inf\{m \geq 1 : X_{md} = k\} = \text{'return time to } k \text{ for } P^d \text{'}$$

Starting from k , we have $T_k^{(d)} = \frac{1}{d}T_k$, so $\mathbb{E}_k[T_k^{(d)}] = \frac{\mathbb{E}_k T_k}{d} < \infty$. Hence P^d is positive recurrent (on each communicating class $C_0, \dots, C_{d-1}$). We can write

$$\mathbb{P}_i[X_{md+r} = j] = \mathbb{P}_w[X_{md} = j],$$

where $w_k = \mathbb{P}_i[X_r = k]$, $k \in I$. Note that $w_k > 0$ only if $k \in C_t$. Applying the convergence theorem to P^d restricted to C_t , we get

$$\mathbb{P}_w[X_{md} = j] \xrightarrow{m \rightarrow \infty} \frac{1}{\mathbb{E}_j T_j^{(d)}} = \frac{d}{\mathbb{E}_j T_j} = d\pi_j.$$

$\square$

Example 1.46 cont'd. We have $\pi = (\frac{5}{14}, \frac{3}{14}, \frac{2}{14}, \frac{4}{14})$.

- If $i, j \in C_0$:

$$\lim_{m \rightarrow \infty} \mathbb{P}_i[X_{2m} = j] = d\pi_j = \begin{cases} 2 \cdot \frac{5}{14} = \frac{10}{14} & \text{when } j = 1; \\ 2 \cdot \frac{2}{14} = \frac{4}{14} & \text{when } j = 3. \end{cases}$$

- If $i, j \in C_1$:

$$\lim_{m \rightarrow \infty} \mathbb{P}_i[X_{2m} = j] = d\pi_j = \begin{cases} 2 \cdot \frac{3}{14} = \frac{6}{14} & \text{when } j = 2; \\ 2 \cdot \frac{4}{14} = \frac{8}{14} & \text{when } j = 4. \end{cases}$$

- If $i \in C_0, j \in C_1$:

$$\lim_{m \rightarrow \infty} \mathbb{P}_i[X_{2m+1} = j] = d\pi_j = \begin{cases} 2 \cdot \frac{3}{14} = \frac{6}{14} & \text{when } j = 2; \\ 2 \cdot \frac{4}{14} = \frac{8}{14} & \text{when } j = 4. \end{cases}$$

- If $i \in C_1, j \in C_0$:

$$\lim_{m \rightarrow \infty} \mathbb{P}_i[X_{2m+1} = j] = d\pi_j = \begin{cases} 2 \cdot \frac{5}{14} = \frac{10}{14} & \text{when } j = 1; \\ 2 \cdot \frac{2}{14} = \frac{4}{14} & \text{when } j = 3. \end{cases}$$

Other terms are 0:

$$\mathbb{P}_i[X_{2m+1} = j] = 0 \quad \text{when } i, j \in C_0; \text{ or if } i, j \in C_1.$$

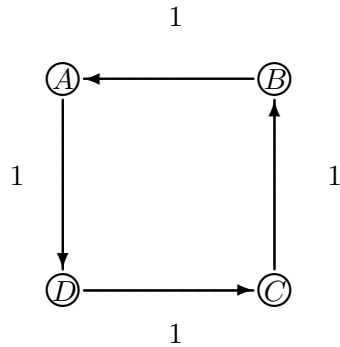
$$\mathbb{P}_i[X_{2m} = j] = 0 \quad \text{when } i \in C_0, j \in C_1; \text{ or if } i \in C_1, j \in C_0.$$

As a special case of Theorem 1.47 we get the so-called Renewal Theorem: If P is irreducible with period $d \geq 1$, and positive recurrent, then for all $i \in I$ we have

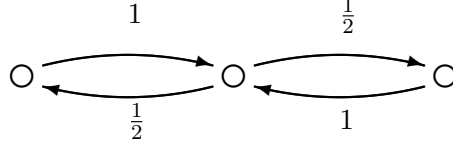
$$\lim_{m \rightarrow \infty} p_{ii}^{(md)} = d\pi_i = \frac{d}{\mathbb{E}_i T_i}.$$

Some further examples follow.

Example 1.48. This is a trivial example without randomness, a particle moves always one step counterclockwise through the graph.



Here the period of every state is 4 and the average return time is also 4, the transition probabilities satisfy $(P^{4n})_{ii} = 1$. The renewal theorem holds (even without the limit!)



Example 1.49. Consider the example given by the following graph.

The one-step transition matrix is given by

$$P = \begin{pmatrix} 0 & 1 & 0 \\ 1/2 & 0 & 1/2 \\ 0 & 1 & 0 \end{pmatrix}.$$

All states have period 2. We can find

- (a) P^n by diagonalization with tricks,
- (b) $\mathbb{E}_i\{T_i\}$ by the “one-step method”.

Then we can verify the statement of the renewal theorem in our example.

(a) We have $\text{trace } P = 0$, $\det P = 0$ and hence eigenvalues $1, 0, -1$. Solving $\pi P = \pi$ gives

$$\pi = (1/4, 1/2, 1/4).$$

Hence

$$P^n = U_1 + (-1)^n U_2 + 0^n U_3 \text{ for all } n \geq 1,$$

and

$$U_1 = \begin{pmatrix} 1/4 & 1/2 & 1/4 \\ 1/4 & 1/2 & 1/4 \\ 1/4 & 1/2 & 1/4 \end{pmatrix},$$

$$U_2 = U_1 - P = \begin{pmatrix} 1/4 & -1/2 & 1/4 \\ -1/4 & 1/2 & -1/4 \\ 1/4 & -1/2 & 1/4 \end{pmatrix},$$

and U_3 is irrelevant. For example, we get

$$(P^n)_{33} = \frac{1}{4} + (-1)^n \frac{1}{4} \text{ for all } n \geq 1.$$

(b) For $y_i = \mathbb{E}_i\{T_3\}$ we get the equations

$$\begin{aligned} y_1 &= 1 + y_2 \\ y_2 &= 1 + \frac{1}{2}y_1 \\ y_3 &= 1 + y_2. \end{aligned}$$

Solving gives, for example, $\mathbb{E}_3\{T_3\} = y_3 = 4$, $\mathbb{E}_1\{T_1\} = 4$ by symmetry and trivially $\mathbb{E}_2\{T_2\} = 2$.

Checking the renewal theorem we observe that $(P^n)_{33} = 0$ if n is odd and

$$\lim_{n \rightarrow \infty} (P^{2n})_{33} = \lim_{n \rightarrow \infty} \frac{1}{4} + (-1)^{2n} \frac{1}{4} = \frac{1}{2} = \frac{d(2)}{\mathbb{E}_3\{T_3\}}.$$

Theorem 1.50 (Ergodic Theorem). *Let P be irreducible and positive recurrent. For any initial distribution w we have*

$$\frac{1}{n} \# \{0 \leq t < n : X_t = j\} \xrightarrow{n \rightarrow \infty} \pi_j \quad \text{with probability 1.}$$

More generally, if $f : I \rightarrow \mathbb{R}$ is bounded (there exists M such that $|f(i)| \leq M$ for all $i \in I$), then

$$\frac{1}{n} \sum_{t=0}^{n-1} f(X_t) \xrightarrow{n \rightarrow \infty} \sum_{j \in I} f(j) \pi_j \quad \text{with probability 1.}$$

Proof. (Non-examinable) See [Theorem 1.10.2 in Norris: Markov Chains]. □

Example 1.46 cont'd.

Suppose there is cost $c(i)$ associated with state i . Then the long-run average cost per step is given by the Ergodic Theorem as:

$$\lim_{n \rightarrow \infty} \frac{1}{n} \sum_{t=0}^{n-1} c(X_t) \stackrel{\text{Ergodic Thm}}{=} \sum_{j \in I} c(j) \pi_j = c(1) \frac{5}{14} + c(2) \frac{3}{14} + c(3) \frac{2}{14} + c(4) \frac{4}{14}.$$

1.6 Time-reversible Markov chains

We discuss the case of time-reversible or symmetrizable Markov chains. Suppose $(m_i : i \in I)$ is a collection of nonnegative numbers, not all zero. We call a Markov chain **m -symmetrizable** if we have

$$m_i p_{ij} = m_j p_{ji} \quad \text{for all } i, j \in I.$$

Lemma 1.51. *If P is m -symmetrizable, and $M := \sum_{i \in I} m_i < \infty$, then $\pi_i := \frac{m_i}{M}$ is an invariant distribution.*

Proof. It is clear that $\sum_{i \in I} \pi_i = 1$, so π is a distribution. We have

$$\pi_i p_{ij} = \pi_j p_{ji} \quad \text{for all } i, j \in I. \tag{1.6.1}$$

This implies

$$\sum_{i \in I} \pi_i p_{ij} = \sum_{i \in I} \pi_j p_{ji} = \pi_j \sum_{i \in I} p_{ji} = \pi_j \quad \text{for all } j \in I.$$

Therefore, π is invariant. □

When (1.6.1) holds, we say that π and P are in **detailed balance**. We call (1.6.1) the **detailed balance equations**.

Example 1.52. Let P be the transition matrix of a simple random walk on the finite graph $G = (V, E)$. We show that P is m -symmetrizable with $m_v = \deg(v)$, $v \in V$. Consequently, $\pi_v = \frac{\deg(v)}{2|E|}$ is an invariant distribution.

Indeed, we have $p_{vw} = \frac{a_{vw}}{\deg(v)}$, where $a_{vw} = \#$ edges between v and w . Then

$$\begin{aligned} m_v p_{vw} &= \deg(v) \frac{a_{vw}}{\deg(v)} = a_{vw} \\ m_w p_{wv} &= \deg(w) \frac{a_{wv}}{\deg(w)} = a_{wv} = a_{vw}. \end{aligned}$$

We have $M = \sum_{v \in V} \deg(v) = 2|E|$, so $\pi_v = \frac{\deg(v)}{2|E|}$, $v \in V$.

Recall Examples 1.44 and 1.46 in the previous section.

We next show that under the law $\mathbb{P}_\pi$ the sequence $X_0, \dots, X_n$ has the same distribution as the time-reversed sequence $X_n, \dots, X_0$.

Proposition 1.53 (Time-reversibility). *If π and P are in detailed balance, then*

$$\mathbb{P}_\pi[X_n = i_0, \dots, X_0 = i_n] = \mathbb{P}_\pi[X_0 = i_0, \dots, X_n = i_n]. \quad (1.6.2)$$

Proof.

$$\begin{aligned} \text{LHS} &= \pi_{i_n} p_{i_n i_{n-1}} p_{i_{n-1} i_{n-2}} \cdots p_{i_2 i_1} p_{i_1 i_0} \\ &\stackrel{(1.6.1)}{=} p_{i_{n-1} i_n} \pi_{i_{n-1}} p_{i_{n-1} i_{n-2}} \cdots p_{i_2 i_1} p_{i_1 i_0} \\ &\stackrel{(1.6.1)}{=} p_{i_{n-1} i_n} p_{i_{n-2} i_{n-1}} \pi_{i_{n-2}} \cdots p_{i_2 i_1} p_{i_1 i_0} \\ &\vdots \\ &\stackrel{(1.6.1)}{=} p_{i_{n-1} i_n} p_{i_{n-2} i_{n-1}} \pi_{i_{n-2}} \cdots p_{i_1 i_2} \pi_{i_1} p_{i_1 i_0} \\ &\stackrel{(1.6.1)}{=} p_{i_{n-1} i_n} p_{i_{n-2} i_{n-1}} \pi_{i_{n-2}} \cdots p_{i_1 i_2} p_{i_0 i_1} \pi_{i_0} = \text{RHS}. \end{aligned}$$

□

When (1.6.1) (and consequently) (1.6.2) hold, we call the Markov chain **reversible** with respect to π .

Remarks:

- If X is π -symmetrizable, then π is an invariant distribution of X . But conversely, if π is an invariant distribution of X this does *not* imply that X is π -symmetrizable.
- It is sometimes much easier to solve the detailed balance equations (1.6.1) and thus find an invariant distribution, rather than solving $\pi P = \pi$, see Example 1.52 above, for instance.
- If the invariant distribution does not solve the detailed balance equations, then they have no solution.

Example 1.54. Consider a simple random walk on $I = \{0, 1, 2, \dots\}$. We attempt to look for m_i such that $m_i p_{i+1} = m_{i+1} p_{i+1}$ for all $i \geq 0$. We get the equations:

$$\begin{aligned} m_0 p &= m_1 (1-p) \implies m_1 = m_0 \frac{p}{1-p} \\ m_1 p &= m_2 (1-p) \implies m_2 = m_1 \frac{p}{1-p} = m_0 \left(\frac{p}{1-p} \right)^2 \\ &\vdots \\ m_i &= m_0 \left(\frac{p}{1-p} \right)^i, \quad i \geq 0. \end{aligned}$$

For what p will be $M < \infty$? We have

$$M = m_0 \sum_{i=0}^{\infty} \left(\frac{p}{1-p} \right)^i = \begin{cases} m_0 \frac{1}{1-\frac{p}{1-p}} < \infty & \text{if } \frac{p}{1-p} < 1 \iff p < 1/2; \\ \infty & \text{if } \frac{p}{1-p} \geq 1 \iff p \geq 1/2. \end{cases}$$

When $p < 1/2$, we have the invariant distribution

$$\pi_i = \frac{m_0 \left(\frac{p}{1-p} \right)^i}{m_0 \frac{1}{1-\frac{p}{1-p}}} = \left(1 - \frac{p}{1-p} \right) \left(\frac{p}{1-p} \right)^i, \quad i \geq 0,$$

which is a $\text{Geom}(1 - \frac{p}{1-p})$ distribution shifted by -1 . In particular, the Markov chain is positive recurrent. As an application, consider the long-run average position of the walker. The Ergodic Theorem suggests (taking $f(i) = i$) that we should have:

$$\text{long-run average position of the walker} = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{t=0}^{n-1} X_t = \sum_{i=0}^{\infty} i \pi_i = \frac{\frac{p}{1-p}}{1 - \frac{p}{1-p}} = \frac{p}{1-2p}.$$

We used that the mean of the shifted $\text{Geom}(q)$ random variable is $(1-q)/q$. Notice that the function $f(i) = i$ here is not bounded, and hence Theorem 1.50 does not apply directly. However, the statement is still true, by an appropriate extension of that theorem (non-examinable).

Example 1.55. Consider further the random walk with $p < 1/2$, started from the equilibrium distribution π . Which event is more likely:

$$\{X_0 = 10, X_1 = 9, X_2 = 8, \dots, X_{10} = 0\} \quad \text{or} \quad \{X_0 = 0, X_1 = 1, X_2 = 2, \dots, X_{10} = 10\}?$$

Reversibility (1.6.2) says that starting in equilibrium, the two are equally likely. The probabilities work out to be:

$$\begin{aligned} \pi_{10} (1-p)^{10} &= \left(1 - \frac{p}{1-p} \right) \left(\frac{p}{1-p} \right)^{10} (1-p)^{10} = \left(1 - \frac{p}{1-p} \right) p^{10} \\ \pi_0 p^{10} &= \left(1 - \frac{p}{1-p} \right) p^{10}. \end{aligned}$$

Note: the two events *do not* have the same probability, if the initial states are fixed to be 10 and 0, respectively:

$$\mathbb{P}_{10}[X_0 = 10, \dots, X_{10} = 0] = (1-p)^{10} \neq p^{10} = \mathbb{P}_0[X_0 = 0, \dots, X_{10} = 10].$$

It is crucial for the reversibility property that we start in the equilibrium distribution π .

Some further examples follow.

Example 1.56. Let X be a Markov chain with state space $I = \{0, 1, 2\}$ and transition matrix

$$P = \begin{pmatrix} 1/2 & 1/4 & 1/4 \\ 1/2 & 1/6 & 1/3 \\ 1/4 & 1/6 & 7/12 \end{pmatrix}$$

- Find the equilibrium distribution π .
- Is P π -symmetrizable?

π has to satisfy $\pi_0 + \pi_1 + \pi_2 = 1$ and, from $\pi P = \pi$,

$$\begin{aligned} \frac{1}{2}\pi_0 + \frac{1}{2}\pi_1 + \frac{1}{4}\pi_2 &= \pi_0, \\ \frac{1}{4}\pi_0 + \frac{1}{6}\pi_1 + \frac{1}{6}\pi_2 &= \pi_1, \\ \frac{1}{4}\pi_0 + \frac{1}{3}\pi_1 + \frac{7}{12}\pi_2 &= \pi_2. \end{aligned}$$

This can be solved to give $\pi = (2/5, 1/5, 2/5)$. To check symmetrizability we have to verify $\pi_i p_{ij} = \pi_j p_{ji}$ for all $i, j \in \{0, 1, 2\}$. There are three non-trivial equations to be checked,

$$\begin{aligned} \pi_0 p_{01} = \pi_1 p_{10} &\Leftrightarrow \pi_0 = 2\pi_1 \\ \pi_0 p_{02} = \pi_2 p_{20} &\Leftrightarrow \pi_2 = \pi_0 \\ \pi_1 p_{12} = \pi_2 p_{21} &\Leftrightarrow \pi_2 = 2\pi_1. \end{aligned}$$

This is satisfied for our π . In fact one could have started with these equations and $\pi_0 + \pi_1 + \pi_2 = 1$ and the only solution is the invariant distribution π .

Example 1.57. Let X be a Markov chain with state space $I = \{0, 1, 2\}$ and transition matrix

$$P = \begin{pmatrix} 1/2 & 1/4 & 1/4 \\ 1/2 & 1/6 & 1/3 \\ 1/2 & 1/6 & 1/3 \end{pmatrix}.$$

This matrix is not symmetrizable. Trying to find a suitable m leads to the equations

$$\begin{aligned} m_0 p_{01} = m_1 p_{10} &\Leftrightarrow m_0 = 2m_1 \\ m_0 p_{02} = m_2 p_{20} &\Leftrightarrow m_0 = 2m_2 \\ m_1 p_{12} = m_2 p_{21} &\Leftrightarrow m_2 = 2m_1. \end{aligned}$$

These equations have only the trivial solution $m_0 = m_1 = m_2 = 0$, which is not permitted in the definition of symmetrizability. Still, one can find an invariant distribution $\pi = (1/2, 5/24, 7/24)$.